

Category of Modules

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Abstract

This essay studies the relationship between category theory and the model theory of modules, particularly focusing on pp conditions and finitely presented functors from $\mathbf{Mod}\text{-}R$ to $\mathbf{Ab}$. The first chapter introduces fundamental concepts, including functors, natural transformations, equivalences, limits and directed colimit, together with foundational results like Yoneda Lemma, equivalence criterion, presentability of $R\text{-}\mathbf{Mod}$.

The second chapter focus on the theory of pp-conditions and pp-pairs. After introducing the basic definitions and properties about pp-conditions and pp-pairs, the essay concludes with the proof of the equivalence of the category of pp-pairs and the category of finitely presented functor from $\mathbf{Mod}\text{-}R$ to $\mathbf{Ab}$, namely Theorem 2.51. This equivalence provides a functorial perspective on the model theory of modules.

Introduction

Category theory, was founded by Samuel Eilenberg and Saunders Mac Lane in the 1940s. It originated from algebraic topology, where Eilenberg and Mac Lane was trying to formalize the idea of “natural” construction in algebraic topology. Mathematicians noticed that many constructions developed in category theory can be found in different context across various field of study. Together with the introduction of adjoint functor by Daniel Kan in 1958, category theory had become a powerful framework to describe natural constructions and relationships between mathematical object. On the other hand, Alexander Grothendieck gave the standard definition of abelian categories in the famous “Tohoku paper” in 1957. This concept provided a general and rigorous setting for homological algebra.

In the field of model theory of modules, a major breakthrough was made in 1976 by Walter Baur, who proved that every first-order formula in the theory of modules, subject to suitable restriction, is equivalent to a Boolean combination of pp-(primitive positive) condition. This fundamental result, which forms the basis of the second chapter of this essay, revealed that the model theoretical structure of modules is captured by pp-condition. As a result, the study of model theory of modules became centred on pp-condition.

A further decisive development came in 1984 with the introduction of Ziegler spectrum by Martin Ziegler. Ziegler had shown that understanding the topological space, ideally having an explicit description of the topology, is crucial for answering model theoretic questions on modules.

In the 90s, Ivo Herzog demonstrated that moving to the suitable functorial perspective is the most natural way to prove Ziegler’s results. In 1997, he proved a general theorem relating pp-pairs and finitely presented functors, which is a generalized version of the last theorem of this essay. Thereby building up the connection between category theory and the model theory of modules. Providing us a functorial perspective in investigating pp-condition.

The first chapter mainly focus on introducing the basics of category theory. In section 1.1 and 1.2, we introduces the basic of category theory, which includes the definition of categories, subcategories, functors, natural transformation and equivalences, together with many examples. The principle of duality and criterion of equivalence (namely a functor is an equivalence if and only if it is fully faithful and essentially surjective on object) is also mentioned. These concepts will be essential later in the essay.

Section 1.3 is devoted to exposit the Yoneda Lemma, which is a fundamental results in category theory and also used in our investigation on

pp-condition.

In section 1.4 and 1.5, we introduces limits and exact sequences which are two fundamental tools in understanding abelian categories and presentable categories. Namely the focus of the next two chapters

Section 1.6 and 1.7 are relatively independent of each other compare with the previous sections. The main goals of these sections are to prove that $R\text{-}\mathbf{Mod}$ is an abelian category and a finitely presentable category respectively. These results ensures us that $R\text{-}\mathbf{Mod}$ is a convenient, natural and well-behaved setting for the investigation of pp-conditions.

The second chapter shifts the focus to the model theory of modules. In section 2.1 and 2.2, the basic model theory of modules and the fundamental notions about pp-conditions and pp-pairs are introduced. Most of the propositions in section 2.2 are mentioned for the purpose of enhancing understanding with pp-conditions. Section 2.3 introduces quiver to provide more examples of pp-conditions.

Finally in section 2.4, we are using the definitions and lemmas established in previous sections to prove the main theorem 2.51.

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1 Category Theory

1.1 Introduction to categories

The development of category theory is driven by the realization that various fields of mathematics can be integrated and characterized using the concept of morphism. Rather than focusing on the internal structure of mathematical objects, category theory emphasizes the relationships between them. This shift in perspective allows seemingly different areas—such as algebra, topology, and logic—to be studied within a unified framework. In this chapter, we will introduce the basics of category theory, mostly relying on the texts Leinster (2014) and Richter (2020).

Definition 1.1

(Leinster (2014) Definition 1.1.1) A category $\mathcal{C}$ consists of the following.

- A collection ¹ called *objects*, denoted as $ob(\mathcal{C})$ or simply as $\mathcal{C}$.
- For each $A, B \in ob(\mathcal{C})$, a collection ¹ of *morphisms* from A to B is denoted as $\mathcal{C}(A, B)$ or $Hom_{\mathcal{A}}(A, B)$.
- For each $A, B, C \in ob(\mathcal{C})$, a function $\circ : \mathcal{C}(B, C) \times \mathcal{C}(A, B) \rightarrow \mathcal{C}(A, C)$ is called composition, where $\circ(g, f)$ is denoted as $g \circ f$.
- For each $A \in ob(\mathcal{C})$, the *identity* morphism $1_A \in \mathcal{C}(A, A)$.

Satisfying the following axioms:

- Associativity: $h \circ (g \circ f) = (h \circ g) \circ f$,
for all $h \in \mathcal{C}(C, D)$, $g \in \mathcal{C}(B, C)$, $f \in \mathcal{C}(A, B)$.
- Identity Laws: For any $A, B \in \mathcal{C}$ and each $f \in \mathcal{C}(A, B)$,

$$f \circ 1_A = f = 1_B \circ f.$$

The definition is general enough to capture different mathematical structures across various fields of study. Informally, a category can be thought of as a network of objects connected by morphisms, where the axioms ensure that these arrows can be composed in a consistent way, mirroring familiar operations like function composition.

Definition 1.2

(Leinster (2014) p.75)

A category $\mathcal{A}$ is called small if $ob(\mathcal{A})$ form a set, otherwise it is a large category.

A category $\mathcal{A}$ is called locally small if $\mathcal{A}(A, B)$ form a set for every $A, B \in ob(\mathcal{A})$.

¹The collection of objects and morphisms can be a *proper class* instead of a *set*. The difference will be neglected in this article.

Notice that any small category is a locally small category. These distinctions between small and large category are mainly set-theoretic in nature and for the purpose of avoiding logical paradoxes. This article will omit discussions about this issue. Most of the category introduce in this paper is locally small.

Example 1.3

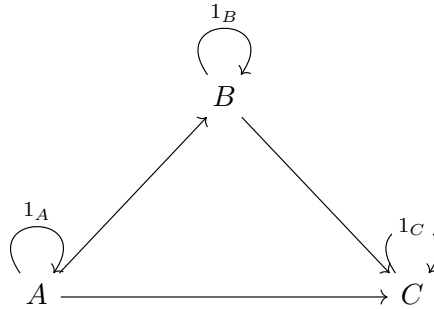
- **Set** is the category of sets where $ob(\mathbf{Set})$ is the collection of all sets and $\mathbf{Set}(A, B)$ is the collection of all functions from set A to set B for each $A, B \in ob(\mathbf{Set})$.
- **Grp** is the category of groups where $ob(\mathbf{Grp})$ is the collection of all groups and $\mathbf{Grp}(A, B)$ is the collection of all group homomorphisms from group A to group B for each $A, B \in ob(\mathbf{Grp})$.
- **0** is the empty category; It contains no objects and no morphisms.
- **1** is the category consisting of exactly one object, and the only morphism in this category is the identity morphism on that object, as illustrated in the figure below:



- **2** is the category with two objects, denoted A and B . Its morphisms are identity morphisms on A and B , together with a non-identity morphism from A to B .



- **3** is the category with three objects, denoted A , B and C . Its morphisms are defined analogously, namely identity morphisms for each object and three non-identity morphisms from A to B , from A to C and from B to C respectively.



Observe that the categories **1**, **2** and **3** corresponds to a linearly ordered set with one, two and three elements respectively.

- Any pre-order $(S, \leq)$ has a corresponding category, where the objects are the elements of S and there is a morphism $a \rightarrow b$ if and only if $a \leq b$ for all $a, b \in S$.
- $R\text{-}\mathbf{Mod}$ is the category of modules over ring R where $ob(R\text{-}\mathbf{Mod})$ is the collection of all modules over ring R and $R\text{-}\mathbf{Mod}(A, B)$ is the collection of all module homomorphisms from module A to module B for each $A, B \in ob(R\text{-}\mathbf{Mod})$. The second part of this essay will mainly focus on the category of modules.
- $\mathbf{Ab}$ is the category of abelian groups where the objects are abelian groups and morphisms are group homomorphisms. (Notice that any homomorphism between two abelian groups must also preserve the commutativity of elements).
- (Riehl (2014) Example 1.1.4) For a ring R , $\mathbf{Mat}_R$ is a category where $ob(\mathbf{Mat}_R)$ is the collection of positive integers and $\mathbf{Mat}_R(n, m)$ is the set of $m \times n$ matrices with values in R for each $n, m \in ob(\mathbf{Mat}_R)$.
- A category $\mathcal{A}$ with a single object A , in which every morphism is an isomorphism, can be regarded as a group via the following correspondence:
 - Each morphism $g \in \mathcal{A}(A, A)$ is identified with an element of the group.
 - The composition map $\circ : \mathcal{A}(A, A) \rightarrow \mathcal{A}(A, A)$ is identified with the group multiplication.
 - The identity morphism 1_A is identified with the identity element of the group.

More generally, a category with a single object whose morphisms are all endomorphisms (not necessarily invertible), can be regarded as a monoid by analogous correspondence.

These examples demonstrate that category theory provides a unifying language for describing mathematical objects across multiple fields by focusing on morphisms and their composition. This reveals structural similarities between different areas of mathematics.

Definition 1.4

(Richter (2020) Definition 1.4.1) Let $\mathcal{C}$ be a category. A subcategory $\mathcal{D}$ of $\mathcal{C}$ is a category such that:

- $ob(\mathcal{D}) \subseteq ob(\mathcal{C})$.
- For all $D_1, D_2 \in ob(\mathcal{D})$, $\mathcal{D}(D_1, D_2) \subseteq \mathcal{C}(D_1, D_2)$.
- If $f \in \mathcal{D}(D_1, D_2)$, then $D_1, D_2 \in ob(\mathcal{D})$.
- For all $D \in \mathcal{D}$, the identity morphism 1_D is in $\mathcal{D}(D, D)$.
- If $f \in \mathcal{D}(D_1, D_2)$, $g \in \mathcal{D}(D_2, D_3)$, then $g \circ f \in \mathcal{D}(D_1, D_3)$.

A subcategory is a concept that imitate subset when considering set, it is obtained by restricting attention to certain objects and morphisms while preserving compatibility of composition and identities with the larger category.

Definition 1.5

For any category $\mathcal{A}$, we can construct $\mathcal{A}^{op}$, called the dual category (or the opposite category) of $\mathcal{A}$, in the following way.

- $ob(\mathcal{A}^{op}) = ob(\mathcal{A})$.
- For all $A, B \in ob(\mathcal{A})$ and all $f \in \mathcal{A}(B, A)$, there is a corresponding $f^{op} \in \mathcal{A}^{op}(A, B)$.

Besides the construction of dual category, there are many notions that comes in dual pair in category theory. The principle of duality introduce below therefore is greatly beneficial in simplifying arguments, it allows us to avoid unnecessary proof for the dual statement and reveals a fundamental symmetry inherent in category theory.

Theorem 1.6

(Awodey (2006) Proposition 3.1)

Any definition or theorem can be **dualised** by reversing the arrow. To put it more formally:

We define the elementary language of category theory consists of objects: $A, B, C, \dots$; arrows: $f, g, h, \dots$ and operation $dom(f)$, $cod(f)$, 1_A , $\circ$ which represent domain, codomain, identity and composition respectively. Note that $g \circ f$ is only defined when $dom(g) = cod(f)$.

Together with seven axioms to follow:

- $dom(1_A) = A$.
- $cod(1_A) = A$.

- $f \circ 1_{\text{dom}(f)} = f$.
- $1_{\text{cod}(f)} \circ f = f$.
- $\text{dom}(g \circ f) = \text{dom}(f)$.
- $\text{cod}(g \circ f) = \text{cod}(g)$.
- $h \circ (g \circ f) = (h \circ g) \circ f$.

Given any statement Σ in the elementary language of category theory, we can form a dual statement Σ^* with respective replacement. Explicitly, it means to replace $f \circ g$ for $g \circ f$; cod for dom ; dom for cod .

The principle of duality asserts that if Σ follows from the axioms, then so does Σ^* . Notice that each axiom beside associativity are dual statement of a corresponding axiom, this is why the principle of duality holds.

In practice, we will invoke the principle of duality without explicit explanation or formal justification if such statement is standard and widely understood.

Definition 1.7

- An epimorphism is a morphism that is right cancellable (i.e. f is an epimorphism if and only if $g \circ f = h \circ f \implies g = h$ for all functions g, h with respective domain and codomain).
- A monomorphism is a morphism whose dual (i.e. the corresponding morphism in dual category) is an epimorphism. In other words, a monomorphism is a morphism that is left cancellable (i.e. m is a monomorphism if and only if $m \circ k = m \circ p \implies k = p$ for all functions k, p with respective domain and codomain).
- An isomorphism is a morphism $f : A \rightarrow B$ in which there exists a morphism $g : B \rightarrow A$ such that $g \circ f = 1_A$ and $f \circ g = 1_B$ where 1_A and 1_B are the identity of objects A and B respectively. We say A and B are isomorphic, denoted as $A \cong B$ if there exist an isomorphism between A and B .

Example 1.8

- (Leinster (2014) Example 5.2.18) The epimorphisms and monomorphisms in the category of sets **Set** are exactly surjective functions and injective functions, respectively.

To prove the statement for epimorphisms, namely a morphism in the category of sets is an epimorphism if and only if it is a surjective function. Observe first that any surjective function is an epimorphism, which

follows directly from elementary set-theoretic considerations.

Conversely, let $f : A \rightarrow B$ be an epimorphism. If B is empty set, then f is trivially surjective. Otherwise, suppose for the sake of contradiction that f is not surjective. one can then define a set $C = \{c_0, c_1\}$ and functions $g, h : B \rightarrow C$ by

$$g(b) = \begin{cases} c_0 & b \in f(A) \\ c_1 & b \in B \setminus f(A) \end{cases}$$

and $h(b) = c_0$ for all $b \in B$. Then $g \circ f = h \circ f$ but $g \neq h$, contradicting the assumption that f is an epimorphism.

To prove the statement for monomorphisms, observe first that any injective function is an monomorphism, which follows directly from elementary set-theoretic considerations.

For the other direction, let $m : B \rightarrow C$ be a monomorphism. If B is empty set or singleton set, then m is trivially injective. Otherwise, suppose $m(b_0) = m(b_1)$ for some $b_0, b_1 \in B$. One can define $k, p : A \rightarrow B$ for some non-empty set A by $k(a) = b_0$ and $p(a) = b_1$ for all $a \in A$. In other words, k and p are the constant function for b_0 and b_1 respectively. Then $m \circ k(a) = m(b_0) = m(b_1) = m \circ p(a)$ for all $a \in A$. By the definition of monomorphism, we have that $k = p$, which means that $b_0 = b_1$, and thus m is injective.

- (Leinster (2014) Example 5.2.19) The epimorphisms and monomorphisms in the categories **Grp**, **Vect_k**, **Top**, **R-Mod** are surjective (group, vector space, continuous, module) homomorphisms and injective (group, vector space, continuous, module) homomorphisms, respectively.
- (Leinster (2014) Exercise 5.2.23) In the category of (unital) rings **Ring**, an epimorphism does not imply Surjectivity. For example:

$$\mathbb{Z} \xhookrightarrow{i} \mathbb{Q}$$

is the inclusion map from $\mathbb{Z}$ to $\mathbb{Q}$, which is right cancellable but not surjective. To see that the map is right cancellable, consider any two ring homomorphisms $g, h : \mathbb{Q} \rightarrow R$ for some ring R . For any $n \in \mathbb{Z}$, if $g \circ i(n) = h \circ i(n)$, then $g(n) = h(n)$ by the definition of inclusion map. For any $q \in \mathbb{Q}$, let $q = \frac{m}{n}$ for some $m, n \in \mathbb{Z}$. By the definition of ring homomorphism, $g(q) = g(m)g(n^{-1}) = g(m)(g(n))^{-1} = h(m)(h(n))^{-1} = h(q)$. Hence the inclusion map is an epimorphism.

- For any category induced by a poset mentioned in Example 1.3. Since any morphisms with the same domain and codomain are the same, this implies that all morphisms are both monomorphisms and epimorphisms.

1.2 Functors and Natural Transformations

This section is focus on introducing the definition of functors and natural transformations, which is fundamental for studying relationship between categories.

Definition 1.9

(Leinster (2014) Definition 1.2.1) A functor $F : \mathcal{A} \rightarrow \mathcal{B}$ is composed of

- A function
 $F : ob(\mathcal{A}) \rightarrow ob(\mathcal{B})$
denoted as $A \mapsto F(A)$.
- For each pair of $A, A' \in \mathcal{A}$, a function
 $F : \mathcal{A}(A, A') \rightarrow \mathcal{B}(F(A), F(A'))$
denoted as $f \mapsto F(f)$.

Satisfying the following axioms:

- $F(1_A) = 1_{F(A)}$ for all objects $A \in \mathcal{A}$.
- $F(g \circ f) = F(g) \circ F(f)$ for all morphisms $f \in \mathcal{A}(A, B), g \in \mathcal{A}(B, C)$.

A functor may be viewed as a structure preserving correspondence between categories, namely it assigns objects and morphisms in a manner compatible with identities and composition.

Example 1.10

- A forgetful functor is a functor where it forgets the algebraic structure of the object and morphism. For example $F : \mathbf{Grp} \rightarrow \mathbf{Set}$ sends a group to its underlying set, and sends a group homomorphism to its underlying function.
- A functor from the category $\mathbf{1}$ to any category $\mathcal{A}$, can be thought of an object $A \in ob(\mathcal{A})$, where A is where the functor maps the unique element in $\mathbf{1}$ to, and the only morphism in $\mathbf{1}$ which is the identity will trivially maps to the identity of A .
- For any two pre-ordered sets, we can construct two categories as shown in Example 1.3. A functor between those two categories corresponds to an order-preserving function between those two pre-ordered sets.

Besides the definition of functor, we also want to describe “morphism” between two functors.

Definition 1.11

(Richter (2020) Definition 2.1.1)

For two functors F, G from $\mathcal{C}$ to $\mathcal{D}$, a natural transformation η from F to G is a class of morphisms $\eta_C \in \mathcal{D}(F(C), G(C))$ for $C \in \text{ob}(\mathcal{C})$, called the component of η at object C , such that for each morphism $f \in \mathcal{C}(C_1, C_2)$:

$$\eta_{C_2} \circ F(f) = G(f) \circ \eta_{C_1}$$

In other words, the following diagram commutes.

$$\begin{array}{ccc} F(C_1) & \xrightarrow{F(f)} & F(C_2) \\ \downarrow \eta_{C_1} & & \downarrow \eta_{C_2} \\ G(C_1) & \xrightarrow{G(f)} & G(C_2) \end{array}$$

Example 1.12

(Leinster (2014) Example 1.3.5) A familiar example of natural transformation is the determinant of a matrix. Consider the forgetful functor from the category of commutative rings to the category of monoids:

$$U : \mathbf{CRing} \rightarrow \mathbf{Mon}$$

It sends a commutative ring R to its underlying monoid under multiplication $U(R)$; Let

$$M_n : \mathbf{CRing} \rightarrow \mathbf{Mon}$$

denote the functor that sends a commutative ring R to the multiplicative monoid of $n \times n$ matrices over R under matrix multiplication.

A family of monoid homomorphisms, denoted by $\det = \{\det_R\}_{R \in \text{ob}(\mathbf{CRing})}$, where for each $R \in \text{ob}(\mathbf{CRing})$, the map $\det_R : M_n(R) \rightarrow U(R)$ is the determinant function, which sends a matrix to its determinant. This function is a component of $\det$ on the object R . And $\det$ satisfying the naturality axiom reflects that the determinant map is defined in the same way for all rings.

$$\begin{array}{ccc} & \xrightarrow{M_n} & \\ \mathbf{CRing} & \begin{array}{c} \Downarrow \det \\ \Downarrow U \end{array} & \mathbf{Mon} \\ & \xrightarrow{U} & \end{array}$$

The naturality square is as follows:

$$\begin{array}{ccc} M_n(R) & \xrightarrow{M_n(f)} & M_n(R') \\ \downarrow \det_R & & \downarrow \det_{R'} \\ U(R) & \xrightarrow{U(f)} & U(R') \end{array}$$

To be more concrete, consider the commutative rings $\mathbb{Z}$ and $\mathbb{F}_7 = \mathbb{Z}/7\mathbb{Z}$, the map $f : \mathbb{Z} \rightarrow \mathbb{F}_7$ which sends an integer to its equivalence class modulo 7, and the matrix $\begin{pmatrix} 4 & 9 \\ 2 & 7 \end{pmatrix}$.

$$U(f)(\det_{\mathbb{Z}}\left(\begin{pmatrix} 4 & 9 \\ 2 & 7 \end{pmatrix}\right)) = U(f)(10) = 3$$

$$\det_{\mathbb{F}_7}(M_n(f)\left(\begin{pmatrix} 4 & 9 \\ 2 & 7 \end{pmatrix}\right)) = \det_{\mathbb{F}_7}\left(\begin{pmatrix} 4 & 2 \\ 2 & 0 \end{pmatrix}\right) = -4 \equiv 3 \pmod{7}$$

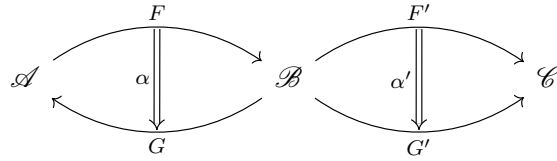
Thus you can see the naturality axiom is satisfied in this example.

Definition 1.13

Given functors $F, G, H : \mathcal{A} \rightarrow \mathcal{B}$ and natural transformations $\alpha : F \rightarrow G$; $\beta : G \rightarrow H$, the composition of the natural transformation is defined component wise as: $(\beta \circ \alpha)_A = \beta_A \circ \alpha_A$.

Remark 1.14

(Leinster (2014) Remark 1.3.24) There is also horizontal composition for natural transformations. Consider functors $F, G : \mathcal{A} \rightarrow \mathcal{B}$; $F', G' : \mathcal{B} \rightarrow \mathcal{C}$ and natural transformations $\alpha : F \rightarrow G$, $\alpha' : F' \rightarrow G'$.



We define $\alpha' * \alpha : F' \circ F \rightarrow G' \circ G$ component wise as $(\alpha' * \alpha)_A = \alpha'_{G(A)} \circ F'(\alpha_A) = G'(\alpha_A) \circ \alpha'_{F(A)}$ for each $A \in \mathcal{A}$.

Example 1.15

A functor category can be constructed after defining the composition for natural transformations. Given two categories $\mathcal{A}, \mathcal{B}$, the functor category $[\mathcal{A}, \mathcal{B}]$ is defined as follows:

- $ob([\mathcal{A}, \mathcal{B}])$ is the collection of all functors from $\mathcal{A}$ to $\mathcal{B}$.
- For $F, G : \mathcal{A} \rightarrow \mathcal{B}$, $[\mathcal{A}, \mathcal{B}](F, G)$ is the collection of natural transformations from F to G .
- For $F : \mathcal{A} \rightarrow \mathcal{B}$, $\mathbf{1}_F$ is the identity natural transformation from F to F such that each component of $\mathbf{1}_F$ at object $x \in ob(\mathcal{A})$ is the identity morphism $1_{F(x)}$.

Definition 1.16

(Leinster (2014) Definition 1.3.10) A natural isomorphism between functors is an isomorphism in the functor category $[\mathcal{A}, \mathcal{B}]$.

Lemma 1.17

(Leinster (2014) Lemma 1.3.11) A natural transformation is natural isomorphism if and only if all of its components are isomorphisms.

Proof.

($\implies$)

Let η be the natural isomorphism from F to G for $F, G : \mathcal{A} \rightarrow \mathcal{B}$. Hence there exist η^{-1} from G to F such that $\eta \circ \eta^{-1} = \mathbf{1}_F$ and $\eta^{-1} \circ \eta = \mathbf{1}_G$ where $\mathbf{1}$ is the identity natural transformation.

For any object $A \in ob(\mathcal{A})$,

$$(\eta^{-1} \circ \eta)_A = \eta_A^{-1} \circ \eta_A = \mathbf{1}_{F(A)},$$

$$(\eta \circ \eta^{-1})_A = \eta_A \circ \eta_A^{-1} = \mathbf{1}_{G(A)},$$

which implies that each component is an isomorphism.

($\impliedby$)

Let λ be a natural transformation such that all of its components are isomorphism. Let λ_A be a component of λ , we have λ_A^{-1} as the inverse. It can be easily checked that the family of λ_A^{-1} forms a natural transformation λ^{-1} such that it is the inverse of λ in the functor category. $\square$

Definition 1.18

(Leinster (2014) Definition 1.3.15) $\mathcal{A}$ and $\mathcal{B}$ are equivalent if there exists a pair of functors $F : \mathcal{A} \rightarrow \mathcal{B}; G : \mathcal{B} \rightarrow \mathcal{A}$ together with a pair of natural isomorphisms: $\eta : \mathbf{1}_{\mathcal{A}} \rightarrow G \circ F; \lambda : F \circ G \rightarrow \mathbf{1}_{\mathcal{B}}$. The functors F and G are called equivalences.

Definition 1.19

(Leinster (2014) Definition 1.2.16, 1.3.17)

- A functor $F : \mathcal{A} \rightarrow \mathcal{B}$ is faithful or full if for each $A, A' \in ob(\mathcal{A})$ and any $f : A \rightarrow A'$ the function $f \mapsto F(f)$ is injective or surjective, respectively.
- A functor F is essentially surjective on objects if for all $B \in ob(\mathcal{B})$, there exists $A \in ob(\mathcal{A})$ such that $F(A) \cong B$.

Theorem 1.20

Leinster (2014) A functor $F : \mathcal{A} \rightarrow \mathcal{B}$ is an equivalence if and only if F is full, faithful, and essentially surjective on objects.

Proof.

(Riehl (2014) Theorem 1.5.9)

($\Leftarrow$)

Let $F : \mathcal{C} \rightarrow \mathcal{D}$ is full, faithful, and essentially surjective. By essential surjectiveness and the axiom of choice, for every $D \in \mathcal{D}$ we can *choose* a $C_D \in \mathcal{C}$ with an isomorphism $\epsilon_D : F(C_D) \rightarrow D$. Consider the following:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \uparrow \epsilon_A & & \uparrow \epsilon_B \\ F(C_A) & \xrightarrow{\epsilon_B^{-1} \circ f \circ \epsilon_A} & F(C_B) \end{array}$$

Since F is full and faithful, there is a unique morphism $u : C_A \rightarrow C_B$ such that $F(u) = \epsilon_B^{-1} \circ f \circ \epsilon_A$.

We want to show that $G(D) = C_D$ and $G(f) = u$ for any $D \in \mathcal{D}$ and any morphism $f \in \text{Hom}(\mathcal{D})$ defines a functor $G : \mathcal{D} \rightarrow \mathcal{C}$. Explicitly, we want to prove that G follows the two equations: $G(1_D) = 1_{G(D)}$ and $G(g \circ f) = G(g) \circ G(f)$.

Consider the following diagram:

$$\begin{array}{ccc} D & \xrightarrow{1_D} & D \\ \uparrow \epsilon_D & & \uparrow \epsilon_D \\ F(G(D)) & \xrightarrow{F(G(1_D))} & F(G(D)) \end{array}$$

We can see from the diagram that,

$$F(G(1_D)) = \epsilon_D^{-1} \circ 1_D \circ \epsilon_D = \epsilon_D^{-1} \circ \epsilon_D = 1_{F(G(D))}.$$

Since $F(1_{G(D)}) = 1_{F(G(D))} = F(G(1_D))$ and F being faithful, we have $G(1_D) = 1_{G(D)}$.

For the composition condition, consider the following diagram.

$$\begin{array}{ccccc}
 A & \xrightarrow{f} & B & \xrightarrow{g} & C \\
 \uparrow \epsilon_A & & \uparrow \epsilon_B & & \uparrow \epsilon_C \\
 F(G(A)) & \xrightarrow{F(G(f))} & F(G(B)) & \xrightarrow{F(G(g))} & F(G(C))
 \end{array}$$

This diagram commutes as it is the combination of two naturality squares of η . Since $f = \epsilon_B \circ F(G(f)) \circ \epsilon_A^{-1}$ and $g = \epsilon_C \circ F(G(g)) \circ \epsilon_B^{-1}$, therefore we have,

$$g \circ f = \epsilon_C \circ F(G(g)) \circ \epsilon_B^{-1} \circ \epsilon_B \circ F(G(f)) \circ \epsilon_A^{-1} = \epsilon_C \circ F(G(g)) \circ F(G(f)) \circ \epsilon_A^{-1}.$$

By the second commutative square as the following.

$$\begin{array}{ccc}
 A & \xrightarrow{g \circ f} & C \\
 \uparrow \epsilon_A & & \uparrow \epsilon_C \\
 F(G(A)) & \xrightarrow{F(G(g \circ f))} & F(G(C))
 \end{array}$$

Thus we have that $\epsilon_C \circ F(G(g \circ f)) \circ \epsilon_A^{-1} = g \circ f = \epsilon_C \circ F(G(g)) \circ F(G(f)) \circ \epsilon_A^{-1}$. Compositing ϵ_C^{-1} on the left and ϵ_A on the right to both side of the equation gives us $F(G(g \circ f)) = F(G(g)) \circ F(G(f))$. By functoriality of F , $F(G(g \circ f)) = F(G(g) \circ G(f))$. By the faithfulness of F , we can conclude that $G(g \circ f) = G(g) \circ G(f)$.

Notice that $\epsilon : F \circ G \rightarrow 1_{\mathcal{D}}$ defined by the collection of isomorphisms ϵ_D for $D \in \mathcal{D}$ is a natural isomorphism, the naturality squares commutes as above had shown. Consider by the full and faithfulness of F , there is a unique morphism $\eta_C : C \rightarrow G(F(C))$ such that $F(\eta_C) = \epsilon_{F(C)}^{-1}$. The result of η_C being isomorphism for all $C \in \mathcal{C}$ immediately follows from the fact that the inverse is the canonical choice : $\epsilon_{F(C)}$. It remains to prove that the collection of η_C forms a natural transformation, namely if the following diagram commutes.

$$\begin{array}{ccc}
 G(F(C)) & \xrightarrow{G(F(f))} & G(F(C')) \\
 \uparrow \eta_C & & \uparrow \eta_{C'} \\
 C & \xrightarrow{f} & C'
 \end{array}$$

In order to prove the naturality of η , we need to consider another diagram:

$$\begin{array}{ccc}
F(G(F(C))) & \xrightarrow{F(G(F(f)))} & F(G(F(C'))) \\
\uparrow F(\eta_C)=\epsilon_{F(C)}^{-1} & & \uparrow F(\eta_{C'})=\epsilon_{F(C')}^{-1} \\
F(C) & \xrightarrow{F(f)} & F(C')
\end{array}$$

By the naturality of ϵ , this diagram commutes, and thus we have

$$F(G(F(f))) \circ F(\eta_C) = F(\eta_{C'}) \circ F(f).$$

By the faithfulness of F , we get

$$G(F(f)) \circ \eta_C = \eta_{C'} \circ f,$$

which is exactly the naturality axioms for η .

($\implies$)

We will first prove that F is essentially surjective.

Let $G : \mathcal{B} \rightarrow \mathcal{A}$ be the functor such that $\eta : 1_{\mathcal{A}} \rightarrow G \circ F; \lambda : F \circ G \rightarrow 1_{\mathcal{B}}$ be the pair of natural isomorphism as stated in the definition.

Consider $B \in ob(\mathcal{B})$, we have $F(G(B)) = F \circ G(B) \cong B$ by λ_B , and $G(B) \in ob(\mathcal{A})$. Hence F is essentially surjective on objects.

For faithfulness, for any $f, g : A \rightarrow B$ with $A, B \in ob(\mathcal{A})$, we want to show that $F(f) = F(g) \implies f = g$. If $F(f) = F(g)$ then we have $G \circ F(f) = G \circ F(g)$.

Consider the following:

$$\begin{array}{ccc}
G \circ F(A) & \xrightarrow{G \circ F(f)=G \circ F(g)} & G \circ F(B) \\
\uparrow \eta_A & & \uparrow \eta_B \\
A & \xrightleftharpoons[f]{g} & B
\end{array}$$

By the naturality axiom we have the following equation:

$$\eta_B \circ f \circ \eta_A^{-1} = G \circ F(f) = G \circ F(g) = \eta_B \circ g \circ \eta_A^{-1}.$$

Hence $f = \eta_A \circ G \circ F(f) \circ \eta_B^{-1} = \eta_A \circ G \circ F(g) \circ \eta_B^{-1} = g$.

To prove fullness, we will need to show that for all $g : F(A) \rightarrow F(B)$, there exists a $f : A \rightarrow B$ such that $F(f) = g$.

First we prove a useful equality. Consider the following diagram.

$$\begin{array}{ccc}
 A & \xrightarrow{\eta_A} & G \circ F(A) \\
 \eta_A \downarrow & \nearrow 1_{G \circ F(A)} & \downarrow \eta_{G \circ F(A)} \\
 G \circ F(A) & \xrightarrow{G \circ F(\eta_A)} & G \circ F \circ G \circ F(A)
 \end{array}$$

The upper triangle is constructed by repeating $A \xrightarrow{\eta_A} G \circ F(A)$, and the diagonal morphism is the identity. The whole rectangle commutes due to the naturality of η . This gives us the equation $\eta_A \circ G \circ F(\eta_A) = \eta_A \circ \eta_{G \circ F(A)}$, and compositing η_A^{-1} on both side shows that:

$$G \circ F(\eta_A) = \eta_{G \circ F(A)}. \quad (1)$$

Now, we consider the following diagram.

$$\begin{array}{ccc}
 G \circ F(A) & \begin{array}{c} \xrightarrow{G(g)} \\ \xrightarrow{G \circ F(f)} \end{array} & G \circ F(B) \\
 \eta_A \uparrow & & \uparrow \eta_B \\
 A & \xrightarrow{\quad f = \eta_B^{-1} \circ G(g) \circ \eta_A \quad} & B
 \end{array}$$

where $f = \eta_B^{-1} \circ G(g) \circ \eta_A$ is a map formed by composing the arrow in the graph. We want to show $g = F(f)$, as it is the canonical construction. Notice that proving $G(g) = G \circ F(f)$ is suffice for the conclusion because of the faithfulness of G (G is also an equivalence).

Consider $G \circ F(f)$, the equation

$$G \circ F(f) = (G \circ F(\eta_B^{-1})) \circ (G \circ F \circ G(g)) \circ (G \circ F(\eta_A))$$

is given by substituting $f = \eta_B^{-1} \circ G(g) \circ \eta_A$ and applying the functorality of $G \circ F$. Then by the previous identity (1), we have:

$$G \circ F(f) = (\eta_{G \circ F(B)}^{-1}) \circ (G \circ F \circ G(g)) \circ (\eta_{G \circ F(A)}),$$

$$(\eta_{G \circ F(B)}) \circ (G \circ F(f)) = (G \circ F \circ G(g)) \circ (\eta_{G \circ F(A)}).$$

Then consider the naturality square of η :

$$\begin{array}{ccc} G \circ F(A) & \xrightarrow{G(g)} & G \circ F(B) \\ \eta_{G \circ F(A)} \downarrow & & \downarrow \eta_{G \circ F(B)} \\ G \circ F \circ G \circ F(A) & \xrightarrow{G \circ F \circ G(g)} & G \circ F \circ G \circ F(B) \end{array}$$

This naturality square for $G(g)$ asserts that

$$(G \circ F \circ G(g)) \circ (\eta_{G \circ F(A)}) = (\eta_{G \circ F(B)}) \circ G(g).$$

By the equation above, we get:

$$(\eta_{G \circ F(B)}) \circ (G \circ F(f)) = (G \circ F \circ G(g)) \circ (\eta_{G \circ F(A)}) = (\eta_{G \circ F(B)}) \circ G(g).$$

Then composite $\eta_{G \circ F(B)}^{-1}$ to both sides will give us:

$$G \circ F(f) = G(g).$$

The rest follows from this equation. □

Definition 1.21

- An object T is called terminal if there exist an unique map from every object in the category to T .
- An object I is called initial if there exist an unique map from I to every object in the category.
- An object X is a zero object if it is both initial and terminal.

The idea of the initial object is the “source” in a category and the terminal object is the “sink” in the category.

Example 1.22

Here are some examples of initial and terminal objects for some common categories.

- The empty set is initial in **Set**. For any set S , the empty function is the unique map from the empty set to S .

- Any singleton set is terminal in **Set**. For any set S and any singleton set x , the constant function that sends every element in S to x is the unique map.
- The trivial group is initial in **Grp**.
- $\mathbb{Z}$ is initial in **Ring**.
- The category **1** is terminal in the category of categories **CAT**.
- For any pre-ordered set, the initial object and terminal object of the corresponding category (if they exist) are the smallest element and greatest element, respectively.

Lemma 1.23

(Leinster (2014) Lemma 2.1.8) Let I, I' be initial objects of a category, then there exists a unique isomorphism $\phi : I \rightarrow I'$.

Proof.

There exist unique morphisms $f : I \rightarrow I'$ and $f' : I' \rightarrow I$, then we have $f' \circ f : I \rightarrow I$ and $1_I : I \rightarrow I$, but there can only be one morphism from I to I . Therefore $f' \circ f = 1_I$. $f \circ f' = 1_{I'}$ follows from the same reasoning. $\square$

1.3 Yoneda Lemma

This chapter is devoted to present a concise exposition of the Yoneda Lemma, one of the fundamental results of category theory. While the Yoneda Lemma does not play an essential role in the later sections, it is included for the sake of completeness and self-containment.

Definition 1.24

(Leinster (2014) Definition 4.1.1)

Let $\mathcal{A}$ be a locally small category, $A \in \text{ob}(\mathcal{A})$. H^A is defined as follows:

$$H^A = \mathcal{A}(A, -) : \mathcal{A} \rightarrow \mathbf{Set},$$

such that

- For any $B \in \text{ob}(\mathcal{A})$, $H^A(B) = \mathcal{A}(A, B)$.
- For any morphism $B \xrightarrow{g} B'$, $H^A(g) = g \circ - : \mathcal{A}(A, B) \rightarrow \mathcal{A}(A, B')$, such that

$$p \mapsto g \circ p$$

for any $p \in \mathcal{A}(A, B)$.

Definition 1.25

(Leinster (2014) Definition 4.1.3)

Let $\mathcal{A}$ be a locally small category. A functor $X : \mathcal{A} \rightarrow \mathbf{Set}$ is called representable if $X \cong H^A$ (natural isomorphism). A representation of X is a choice of an object $a \in \mathcal{A}$ and an isomorphism between H^A and X .

Definition 1.26

(Leinster (2014) Definition 4.1.15)

Let $\mathcal{A}$ be a locally small category. For $f : A \rightarrow B$ in $\mathcal{A}(A, B)$, $H^f : H^A \rightarrow H^B$ is a natural transformation defined by component as

$$H_C^f : H^A(C) \rightarrow H^B(C),$$

$$q \mapsto q \circ f,$$

for any $q \in H^A(C)$ and any $C \in \mathcal{A}$.

The collection of H^A forms the embedding defined as follows,

$$H^* : \mathcal{A}^{op} \rightarrow [\mathcal{A}, \mathbf{Set}]$$

with $H^*(A) = H^A$ and $H^*(f) = H^f$ for $f \in \mathcal{A}(A, B)$.

Theorem 1.27 (Yoneda Lemma)

Let $\mathcal{A}$ be a locally small category, for any $A \in ob(\mathcal{A})$ and $X \in [\mathcal{A}, \mathbf{Set}]$:

- There exists a bijection $Y(X, A) : [\mathcal{A}, \mathbf{Set}](H^A, X) \rightarrow X(A)$.
- The bijections $(Y_X)_A := Y(X, A)$ are components of a natural transformation $Y_X : [\mathcal{A}, \mathbf{Set}](H^*, X) \rightarrow X$, where

$$[\mathcal{A}, \mathbf{Set}](H^*, X) : \mathcal{A}^{op} \rightarrow [\mathcal{A}, \mathbf{Set}](H^*, X)$$

is a functor such that

$$[\mathcal{A}, \mathbf{Set}](H^*, X)(A) := [\mathcal{A}, \mathbf{Set}](H^A, X)$$

$$[\mathcal{A}, \mathbf{Set}](H^*, X)(f) := - \circ H^f.$$

- The bijections $(Y_A)_X := Y(X, A)$ are also components of a natural transformation $Y_A : [\mathcal{A}, \mathbf{Set}](H^A, -) \rightarrow \epsilon_A$, where

$$[\mathcal{A}, \mathbf{Set}](H^A, -) : [\mathcal{A}, \mathbf{Set}] \rightarrow \mathbf{Set}$$

is a functor such that $[\mathcal{A}, \mathbf{Set}](H^A, -)(X) := [\mathcal{A}, \mathbf{Set}](H^A, X)$

$$[\mathcal{A}, \mathbf{Set}](H^A, -)(\eta) = - \circ \eta,$$

and

$$\epsilon_A : [\mathcal{A}, \mathbf{Set}] \rightarrow \mathbf{Set}$$

is some functor that $\epsilon_A(X) := X(A)$.

Proof.

The proof will be split into 3 parts corresponding to the statement:

- Consider a natural transformation $\eta \in [\mathcal{A}, \mathbf{Set}](H^A, X)$, its components $\eta_C : H^A(C) = \mathcal{A}(A, C) \rightarrow X(C)$ for some $C \in \text{ob}(\mathcal{A})$. Define $Y(X, A)(\eta) := \eta_A(1_A)$. Now we need to check that the function $Y(X, A)$ is bijective.

Consider $f \in \mathcal{A}(A, B)$, $g \in \mathcal{A}(B, C)$ and $x \in X(A)$ for some $B, C \in \text{ob}(\mathcal{A})$. Define a function $\mu(X, A)_{x, B} : \mathcal{A}(A, B) \rightarrow X(B)$ such that:

$$\mu(X, A)_{x, B}(f) := X(f)(x).$$

Then we claim that $\mu(X, A)_x : H^A \rightarrow X$ as a natural transformation from treating $\mu(X, A)_{x, B}$ as the component on B . One can check that the naturality square below does indeed commute:

$$\begin{array}{ccc} H^A(B) = \mathcal{A}(A, B) & \xrightarrow{g \circ -} & H^A(C) = \mathcal{A}(A, C) \\ \downarrow \mu(X, A)_{x, B} & & \downarrow \mu(X, A)_{x, C} \\ X(B) & \xrightarrow{X(g)} & X(C) \end{array}$$

Furthermore, we can induce a bijection $\mu(X, A) : X(A) \rightarrow [\mathcal{A}, \mathbf{Set}](H^A, X)$ by varying x in $\mu(X, A)_x$, namely $\mu(X, A)(a_0) = \mu(X, A)_{a_0}$ for any $a_0 \in X(A)$. We show that $\mu(X, A)$ is the inverse of $Y(X, A)$.

For any $a \in X(A)$

$$\begin{aligned} Y(X, A) \circ \mu(X, A)(a) &= Y(X, A)(\mu(X, A)_a) \\ &= \mu(X, A)_{a, A}(1_A) \\ &= X(1_A)(a) \\ &= 1_{X(A)}(a) \\ &= a \end{aligned}$$

For any $B \in \text{ob}(\mathcal{A})$ and $f : A \rightarrow B$, consider a natural transformation $\psi \in [\mathcal{A}, \mathbf{Set}](H^A, X)$. By the definition of $Y(X, A)$, we can show that:

$$\begin{aligned} \mu(X, A) \circ Y(X, A)(\psi)(B)(f) &= \mu(X, A)(\psi_A(1_A))(B)(f) \\ &= \mu(X, A)_{\psi_A(1_A), B}(f) \\ &= X(f)(\psi_A(1_A)) \end{aligned}$$

Consider the naturality square of ψ :

$$\begin{array}{ccc}
H^A(A) = \mathcal{A}(A, A) & \xrightarrow{f \circ -} & H^A(B) = \mathcal{A}(A, B) \\
\psi_A \downarrow & & \downarrow \psi_B \\
X(A) & \xrightarrow{X(f)} & X(B)
\end{array}$$

We now perform a diagram chase from $1_A \in \mathcal{A}(A, A)$ to $X(B)$. We have that:

$$\begin{aligned}
\psi_B \circ f \circ -(1_A) &= X(f) \circ \psi_A(1_A) \\
\implies \psi_B(f) &= X(f)(\psi_A(1_A)).
\end{aligned}$$

Thus

$$\mu(X, A) \circ Y(X, A)(\psi)(B)(f) = \psi_B(f)$$

for any $\psi \in [\mathcal{A}, \mathbf{Set}](H^A, X)$, any $f \in \mathcal{A}(A, B)$ and any $B \in \mathcal{A}$. Hence we can say that

$$\mu(X, A) \circ Y(X, A)(\psi)(B) = \psi_B$$

and therefore,

$$\mu(X, A) \circ Y(X, A)(\psi) = \psi.$$

- To claim that $Y_X : [\mathcal{A}, \mathbf{Set}](H^*, X) \rightarrow X$ is a natural transformation, we will need that for any $f \in \mathcal{A}(A, B)$, $A, B \in \text{ob}(\mathcal{A})$

$$X(f) \circ (Y_X)_A = (Y_X)_B \circ (- \circ H^f).$$

In other words, the diagram below commutes:

$$\begin{array}{ccc}
[\mathcal{A}, \mathbf{Set}](H^A, X) & \xrightarrow{- \circ H^f} & [\mathcal{A}, \mathbf{Set}](H^B, X) \\
(Y_X)_A \downarrow & & \downarrow (Y_X)_B \\
X(A) & \xrightarrow{X(f)} & X(B)
\end{array}$$

Consider $\eta \in [\mathcal{A}, \mathbf{Set}](H^A, X)$, $(Y_X)_A(\eta) = Y(X, A)(\eta) = \eta_A(1_A)$ as defined above. Then $X(f) \circ (Y_X)_A(\eta) = X(f)(\eta_A(1_A))$.

For the other direction,

$$(Y_X)_B \circ (- \circ H^f)(\eta) = (Y_X)_B((\eta \circ H^f)) = (\eta \circ H^f)_B(1_B)$$

$$\begin{aligned} (\eta \circ H^f)_B(1_B) &= \eta_B((H^f)_B(1_B)) \\ &= \eta_B(1_B \circ f) \\ &= \eta_B(f) \\ &= X(f)(\eta_A(1_A)) \end{aligned}$$

where the last equation holds by the naturality square of η .

- To claim that $Y_A : [\mathcal{A}, \mathbf{Set}](H^A, -) \rightarrow \mathbf{Set}$ is a natural transformation, we will need that for any $X, Z \in [\mathcal{A}, \mathbf{Set}]$, $\theta \in [\mathcal{A}, \mathbf{Set}](X, Z)$

$$\theta_A \circ Y_A(X) = Y_A(Z) \circ (\theta \circ -)$$

$$\begin{array}{ccc} [\mathcal{A}, \mathbf{Set}](H^A, X) & \xrightarrow{\theta \circ -} & [\mathcal{A}, \mathbf{Set}](H^A, Z) \\ \downarrow Y_A(X) & & \downarrow Y_A(Z) \\ X(A) & \xrightarrow{\theta_A} & Z(A) \end{array}$$

Consider $\eta \in [\mathcal{A}, \mathbf{Set}](H^A, X)$,

$$(Y_A)_X(\eta) = Y(X, A)(\eta) = \eta_A(1_A)$$

as defined above. Then $\theta_A \circ Y_A(X)(\eta) = \theta_A \circ \eta_A(1_A)$.

For the other direction:

$$\begin{aligned} Y_A(Z) \circ (\theta \circ -)(\eta) &= Y_A(Z) \circ (\theta \circ \eta) \\ &= (\theta \circ \eta)_A(1_A) \\ &= \theta_A \circ \eta_A(1_A). \end{aligned}$$

Thereby the proof is completed.

□

1.4 Limit and Colimit

Limit and colimit are some of the most important construction in category theory. This chapter is focused on introducing the definitions of products, equalizers and limits, together with some results about their properties.

Definition 1.28

(Richter (2020) Definition 1.3.11)

- An object P in category $\mathcal{C}$ is called projective if for all epimorphisms $f : A \rightarrow B$ and every $p : P \rightarrow B$ there is a $\lambda \in \mathcal{C}(P, A)$ with $f \circ \lambda = p$; i.e.:

$$\begin{array}{ccc} & P & \\ \swarrow \exists \lambda & \downarrow \forall p & \\ A & \xrightarrow{\forall f} & B \end{array}$$

- Dually, an object I is called injective if for all monomorphisms $g : C \rightarrow D$ in $\mathcal{C}$ and all $j : C \rightarrow I$, there is a $\epsilon \in \mathcal{C}(D, I)$; i.e.:

$$\begin{array}{ccc} & I & \\ \nearrow \forall j & \uparrow \exists \epsilon & \\ C & \xrightarrow{\forall g} & D \end{array}$$

Definition 1.29

(Leinster (2014) Definition 5.1.11) Let $\mathcal{A}$ be a category; $X, Y \in ob(\mathcal{A})$ and $s, t : X \rightarrow Y$ as follows:

$$X \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} Y$$

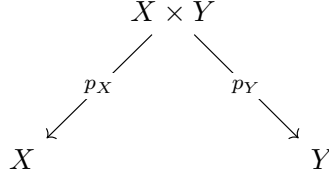
An equalizer of s and t is an object E with map $E \xrightarrow{i} X$ such that $s \circ i = t \circ i$ with the universal property. That is for any $A \xrightarrow{f} X$ that satisfy $s \circ f = t \circ f$, there exist a unique map $A \xrightarrow{f'} E$ such that the following diagram commutes.

$$\begin{array}{ccc} A & & \\ \downarrow f' & \searrow f & \\ E & \xrightarrow{i} & X \end{array}$$

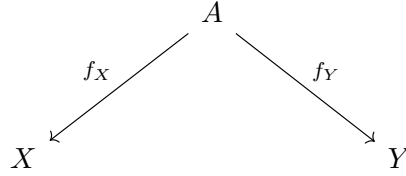
We denote the object E as $eq(s, t)$.

Definition 1.30

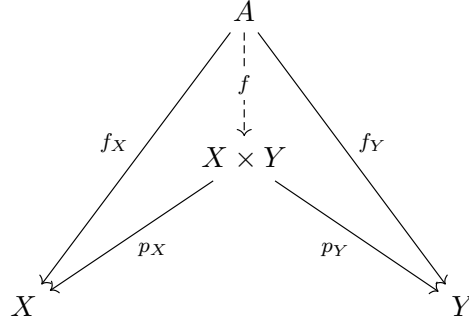
(Leinster (2014) Definition 5.1.1) Let $\mathcal{A}$ be a category and $X, Y \in \text{ob}(\mathcal{A})$. The product of X and Y , denoted as $X \times Y$, is an object in $\mathcal{A}$ with maps $p_X : X \times Y \rightarrow X$ and $p_Y : X \times Y \rightarrow Y$,



such that for any object $A \in \mathcal{A}$ with maps $f_X : A \rightarrow X$ and $f_Y : A \rightarrow Y$,



there exists a unique map $f : A \rightarrow P$ such that the following diagram commutes.



We can also generalize the definition of product, such that the product of a family of objects $(X_i)_{i \in I}$ is defined as the object $\prod_{i \in I} X_i$ with family of maps:

$$(\prod_{i \in I} X_i \xrightarrow{p_j} X_j)_{j \in I}$$

such that for all object A with the family of maps:

$$(A \xrightarrow{f_j} X_j)_{j \in I}$$

there exists a unique map $f : A \rightarrow \prod_{i \in I} X_i$ in which $p_j \circ f = f_j$ for all $j \in I$.

This notion of product is to generalise the cartesian product between set.

Definition 1.31

(Leinster (2014) Definition 5.1.9) Let $\mathcal{D}, \mathcal{I}$ be categories and $F : \mathcal{I} \rightarrow \mathcal{D}$ be a functor, we call the functor F a diagram in $\mathcal{D}$ of shape $\mathcal{I}$.

- A cone of diagram F is an object $D \in \mathcal{D}$ with the family of morphisms:

$$(D \xrightarrow{f_I} F(I))_{I \in \mathcal{I}}$$

such that for all $I \xrightarrow{u} J \in \mathcal{I}$

$$\begin{array}{ccc} & & F(I) \\ & \nearrow f_I & \downarrow F(u) \\ D & & \\ & \searrow f_J & \downarrow \\ & & F(J) \end{array}$$

commutes.

- A limit of diagram F is a cone $(L \xrightarrow{P_I} F(I))_{I \in \mathcal{I}}$ such that for all cones $(C \xrightarrow{f_I} F(I))_{I \in \mathcal{I}}$ there exists a unique map $C \xrightarrow{f} L$ with $P_I \circ f = f_I$ for all $I \in \mathcal{I}$. It is also referred by L when all the morphisms P_I is obvious.

Remark 1.32

A limit $(L \xrightarrow{P_I} F(I))_{I \in \mathcal{I}}$ is also denoted as $\lim_{I \in \mathcal{I}} F(I)$ or $\varprojlim_{I \in \mathcal{I}} F(I)$.

Lemma 1.33

(Leinster (2014) Corollary 6.1.2) Limits are unique up to isomorphism.

Proof.

Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a functor. Let both the cones $(L_1 \xrightarrow{P_I} F(I))_{I \in \mathcal{I}}$ and $(L_2 \xrightarrow{Q_I} F(I))_{I \in \mathcal{I}}$ be limit of diagram F . Then by the definition of limit, there exists a unique map $L_1 \xrightarrow{f} L_2$ and a unique map $L_2 \xrightarrow{g} L_1$ such that $P_I \circ g = Q_I$ and $Q_I \circ f = P_I$. Hence for some $I, J \in \mathcal{I}$, the following diagram commutes.

$$\begin{array}{ccccc} & & L_1 & & \\ & \nearrow P_I & \uparrow & \searrow P_J & \\ & & \vdots & & \\ F(I) & \xleftarrow{\quad} & \vdots & \xrightarrow{\quad} & F(J) \\ & \nwarrow Q_I & \downarrow f & \nearrow Q_J & \\ & & L_2 & & \end{array}$$

Consider $(L_2 \xrightarrow{Q_I} F(I))_{I \in \mathcal{I}}$ as a cone, then by the property of the limit $(L_2 \xrightarrow{Q_I} F(I))_{I \in \mathcal{I}}$, there is a unique map $L_2 \xrightarrow{u} L_2$ such that $Q_I \circ u = Q_I$. Notice that we have $Q_I \circ 1_{L_2} = Q_I = P_I \circ g = (Q_I \circ f) \circ g = Q_I \circ (f \circ g)$ for all $I \in \mathcal{I}$. By the uniqueness of map introduced above, we can conclude that $f \circ g = 1_{L_2}$. Using similar argument on 1_{L_1} and $g \circ f$ will give us $1_{L_1} = g \circ f$. Thus we have that $L_1 \cong L_2$ by the isomorphisms f and g . $\square$

The difference between isomorphic objects is mainly neglected in this essay, we will refer to “the” limit as it is unique.

Definition 1.34

(Richter (2020) Definition 2.6.1)

- A category with all its isomorphic objects being identical is called reduced. In other words, a category $\mathcal{C}$ is reduced if for all objects $A, B \in ob(\mathcal{C})$, we have $A \cong B \implies A = B$.
- $\mathcal{S} \subseteq \mathcal{C}$ is a skeleton if $\mathcal{S}$ is reduced and the inclusion $\mathcal{S} \hookrightarrow \mathcal{C}$ is an equivalence of categories.

Lemma 1.35

Products and equalizers are limits.

Proof.

Let $\mathcal{A}$ be a category, $X, Y \in ob(\mathcal{A})$ and $s, t \in \mathcal{A}(X, Y)$.

For the case of product, we want to show that there exists some functor $H : \mathcal{L} \rightarrow \mathcal{A}$ such that the limit of diagram H is isomorphic to $X \times Y$.

Let $\mathcal{I}$ be the discrete category on two objects, namely the category that consist only of two objects A, B and their identity morphisms $1_A, 1_B$. Now we can construct a functor $F : \mathcal{I} \rightarrow \mathcal{A}$ defined by its action on objects and morphisms as follows: $A \mapsto X, B \mapsto Y; 1_A \mapsto 1_X, 1_B \mapsto 1_Y$. Notice that this definition satisfies the functor axioms immediately as it maps identity to identity and there is no composition of morphisms that is possible.

Let $(L \xrightarrow{\pi_X} X, L \xrightarrow{\pi_Y} Y)$ be the limit of diagram F . Consider the cone $(X \times Y \xrightarrow{p_X} X, X \times Y \xrightarrow{p_Y} Y)$. By the property of $X \times Y$ and the limit L , there is a unique morphism $L \xrightarrow{f} X \times Y$ and a unique morphism $X \times Y \xrightarrow{g} L$ such that $p_A \circ f = \pi_A$ and $\pi_A \circ g = p_A$ for $A = X, Y$. Using similar argument in the proof of Lemma 1.33, we can conclude that $X \times Y \cong L$ with isomorphisms f and g .

For the case of equalizer, we want to show that there exists some functor $H : \mathcal{L} \rightarrow \mathcal{A}$ such that the limit of diagram H is isomorphic to $eq(s, t)$.

Let $\mathcal{J}$ be the category that consists only of two objects A, B , two identities morphisms $1_A, 1_B$, and two morphisms $s', t' \in \mathcal{J}(A, B)$. Consider $G : \mathcal{J} \rightarrow \mathcal{A}$ is the functor that is defined by $A \mapsto X$, $B \mapsto Y$ on objects, and $s' \mapsto s$, $t' \mapsto t$, $1_A \mapsto 1_X$, $1_B \mapsto 1_Y$ on morphisms. One can check easily that this definition also satisfies the functor axioms.

Let $(E \xrightarrow{e_X} X, E \xrightarrow{e_Y} Y)$ be the limit of diagram G . By the property of cone, it follows immediately that $s \circ e_X = e_Y = t \circ e_X$. Consider $eq(s, t)$, by the definition, $(eq(s, t) \xrightarrow{i} X, eq(s, t) \xrightarrow{soi=toi} Y)$ forms a cone and there is a unique morphism $E \xrightarrow{f} eq(s, t)$ such that $i \circ f = e_X$. On the other hand, there is a unique morphism $eq(s, t) \xrightarrow{g} E$ such that $e_X \circ g = i$ and $e_Y \circ g = s \circ i$ by the property of limit. Observe that $i \circ (g \circ f) = i = i \circ 1_{eq(s, t)}$ and $e_X \circ (f \circ g) = e_X = e_X \circ (f \circ g)$, using similar argument in the proof of Lemma 1.33, we can see that $E \cong eq(s, t)$ with isomorphisms f and g . $\square$

Definition 1.36

(Leinster (2014) Definition 5.2.1) Let $\mathcal{D}, \mathcal{J}$ be categories and $F : \mathcal{J} \rightarrow \mathcal{D}$ be a diagram in $\mathcal{D}$. Construct a functor by dualising the functor F and denotes as $F^{op} : \mathcal{J}^{op} \rightarrow \mathcal{D}^{op}$. A cocone on $\mathcal{D}$ is a cone on $\mathcal{D}^{op}$. A colimit of diagram F in $\mathcal{D}$ is a limit of diagram F^{op} in $\mathcal{D}^{op}$.

In particular, a coproduct in $\mathcal{D}$ is a product of $\mathcal{D}^{op}$ and a coequalizer in $\mathcal{D}$ is an equalizer in $\mathcal{D}^{op}$.

A colimit of diagram F is also denoted as $\varinjlim F$.

Remark 1.37

We call a limit small if the source category of the diagram of the limit is a small category.

Theorem 1.38

A category contains all small limits, called small complete if and only if it contains all finite products and equalizers. Dually, a category that contains all small colimits is called small cocomplete if and only if it contains all finite coproducts and coequalizers.

Proof.

(Leinster (2014) Exercise 5.1.38)

We prove the case of a complete category here. The case of cocomplete category follows immediately by the principle of duality.

The implication that a category has all small limit contains all finite products and equalizers follows from the Lemma 1.35. Therefore only the converse is left to prove.

Let $\mathcal{A}$ be a category with all products and equalizers. Let $D : \mathcal{J} \rightarrow \mathcal{A}$ be a small diagram in $\mathcal{A}$. Since the source category $\mathcal{J}$ is small, we can consider the set of objects in $\mathcal{J}$ defined as $Z = \{ K \in \mathcal{J} : \exists u \in \mathcal{J}(J, K) \text{ for some } J \in \mathcal{J} \}$. In other words, Z is the set of objects in $\mathcal{J}$ that is the codomain of some morphism, or to say it is “the collection of the target of some arrows”.

Define two morphisms $s, t : \prod_{I \in \mathcal{J}} D(I) \rightarrow \prod_{K \in Z} D(K)$, where

$$\left(\prod_{I \in \mathcal{J}} D(I) \xrightarrow{p_J} D(J) \right)_{J \in \mathcal{J}}$$

denotes the product of the family of objects $(D(I))_{I \in \mathcal{J}}$. Similarly

$$\left(\prod_{K \in Z} D(K) \xrightarrow{q_V} D(V) \right)_{V \in Z}$$

denotes the product of the family of objects $(D(K))_{K \in Z}$.

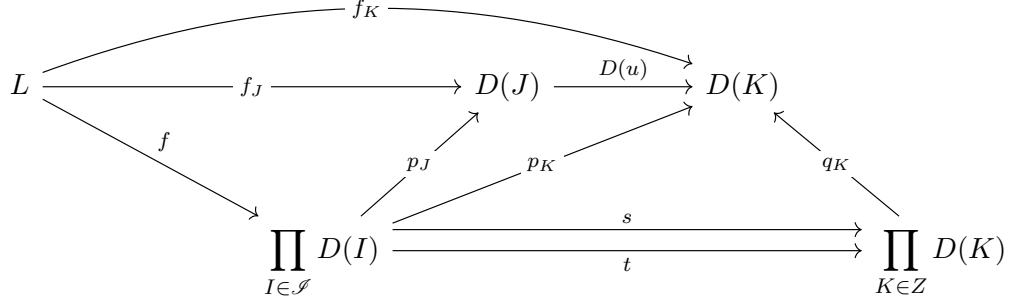
Since a product is determined by its projections, defining the composition $q_K \circ t, q_K \circ s$ for any $K \in Z$ is enough for us to define s and t .

For any $K \in Z$, we can find some $u \in \mathcal{J}(J, K)$ for some $J \in \mathcal{J}$. Define s by $q_K \circ s = D(u) \circ p_J$ for all $K \in Z$. Define t by $q_K \circ t = p_K$. Thus we have the following diagrams commutes.

$$\begin{array}{ccc} & D(J) & \xrightarrow{D(u)} D(K) \\ & \nearrow p_J & \nwarrow q_K \\ \prod_{I \in \mathcal{J}} D(I) & \xrightarrow{s} & \prod_{K \in Z} D(K) \end{array}$$

$$\begin{array}{ccc} & D(K) & \\ & \nearrow p_K & \nwarrow q_K \\ \prod_{I \in \mathcal{J}} D(I) & \xrightarrow{t} & \prod_{K \in Z} D(K) \end{array}$$

Let $L \xrightarrow{f} \prod_{I \in \mathcal{J}} D(I)$ be the equalizer of s and t and denote $f_J = p_J \circ f$ for any $J \in \mathcal{J}$. Explicitly, we will have the following diagram.



Claim: We claim that $(L \xrightarrow{f_I} D(I))_{I \in \mathcal{J}}$ is a limit cone on D .

Proof of the claim:

To prove our claim, we first need to show that $(L \xrightarrow{f_I} D(I))_{I \in \mathcal{J}}$ is a cone on D . Since L is the equalizer of s and t , we have by definition that

$$s \circ f = t \circ f.$$

Consider any $J \xrightarrow{u} K \in \mathcal{J}$, we can composite both side of the equation by q_K :

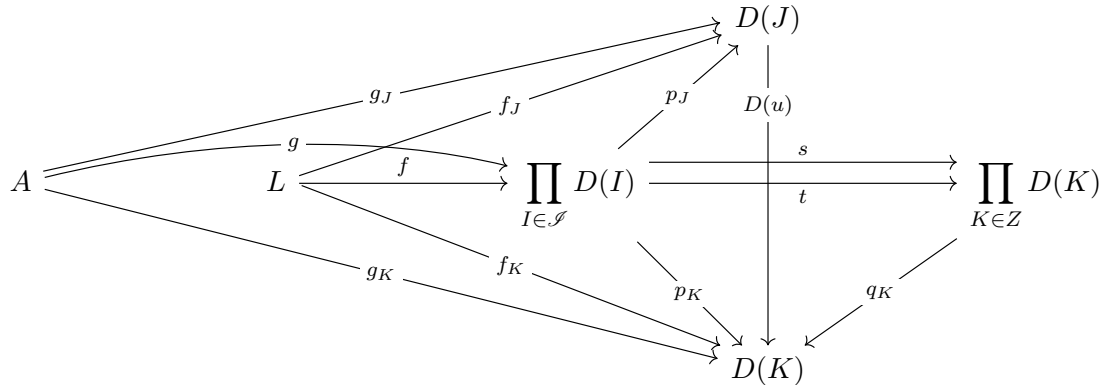
$$\begin{aligned} q_K \circ s \circ f &= q_K \circ t \circ f \\ \implies D(u) \circ p_J \circ f &= p_K \circ f. \end{aligned}$$

By definition of f_K and f_J , we have:

$$D(u) \circ f_J = f_K.$$

Thus we have shown that $(L \xrightarrow{f_I} D(I))_{I \in \mathcal{J}}$ is a cone.

For any cone $(A \xrightarrow{g_I} D(I))_{I \in \mathcal{J}}$, we have a unique map $A \xrightarrow{g} \prod_{I \in \mathcal{J}} D(I)$ such that $p_I \circ g = g_I$ for any $I \in \mathcal{J}$ by the property of the products. Represent by the following diagram.



Observe that $D(u) \circ g_J = g_K$ by the definition of being a cone. Then we have:

$$D(u) \circ p_J \circ g = p_K \circ g.$$

Using the definition of s and t again shows that:

$$q_K \circ s \circ g = q_K \circ t \circ g.$$

Since this is true for any $K \in Z$ and s, t are defined in terms of the composition $q_K \circ s$ and $q_K \circ t$ respectively, thus we have shown that

$$s \circ g = t \circ g.$$

By the property of equalizer, there is a unique map $b : L \rightarrow A$ such that $f \circ b = g$. Then it follows immediately that,

$$f_I \circ b = p_I \circ f \circ b = p_I \circ g = g_I,$$

for any $I \in \mathcal{I}$. Hence proving that $(L \xrightarrow{f_I} D(I))_{I \in \mathcal{I}}$ is a limit cone on D . ■

□

Example 1.39

(Anderson and Fuller (1994) p.79, 82)

Consider the category $R\text{-}\mathbf{Mod}$, a product (categorical definition) of a family of objects $(M_i)_{i \in I}$ for $M_i \in R\text{-}\mathbf{Mod}$ for all $i \in I$, is the direct product of modules M_i , denoted as $\prod_{i \in I} M_i$. On the other hand, a coproduct of a family of objects $(M_i)_{i \in I}$, is the direct sum of modules M_i , denoted as $\bigoplus_{i \in I} M_i$.

The only difference about direct sum and direct product is when the index set I is infinite, then any element in the direct sum, $\vec{x} \in \bigoplus_{i \in I} M_i$, is a tuple with only finitely many non-zero entry, while an element in the direct product can have infinitely many non-zero entry. Explicitly, let $\mathbb{Z}$ be a $\mathbb{Z}$ -module, then $(1, 1, 1, \dots)$ is an element of $\prod_{i \in \mathbb{N}} \mathbb{Z}$, but not an element of $\bigoplus_{i \in \mathbb{N}} \mathbb{Z}$.

Definition 1.40

(Richter (2020) Definition 7.1.6)

Assume that a category $\mathcal{C}$ has a zero object and thus zero morphisms. We define the kernel of a morphism $f \in \mathcal{C}(C, C')$, denoted as $\ker(f)$, as the equalizer of f and the zero morphism $0 \in \mathcal{C}(C, C')$. Explicitly, we mean that $\ker(f) := \text{eq}(f, 0)$. Dually, the cokernel of a morphism f is the coequalizer of f and the zero morphism 0 .

1.5 Abelian Category

In this chapter, we will introduce the notion of abelian category, which provides a natural framework on the study of modules, namely that kernel, cokernel and zero object exists. Notably, $R\text{-Mod}$ for any ring R is an abelian category.

Definition 1.41

(Richter (2020) Definition 7.1.1) A preadditive category is a category $\mathcal{A}$ such that for each pair of objects $A_1, A_2 \in \text{ob}(\mathcal{A})$, $(\mathcal{A}(A_1, A_2), +)$ forms an abelian group for some binary operation $+$, and the composition of the morphisms is a bilinear map with respect to $+$.

Remark 1.42

Notice that a ring R can be regarded as a preadditive category $\mathcal{A}$ with a single object, denoted as $*$, by corresponding the element of the ring to the endomorphisms in $\mathcal{A}$, analogous to Example 1.3. The addition and multiplication are translated to the addition and composition of morphisms respectively.

Definition 1.43

(Richter (2020) Definition 7.1.12)

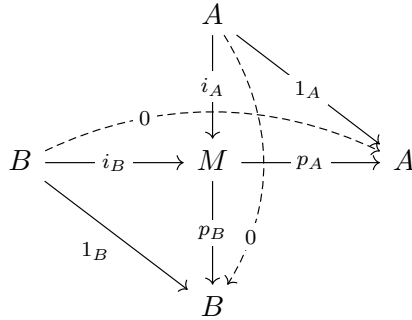
Let $\mathcal{A}$ be a preadditive category. The biproduct of $A, B \in \text{ob}(\mathcal{A})$ is an object $M \in \text{ob}(\mathcal{A})$ with morphisms $p_A : M \rightarrow A$, $p_B : M \rightarrow B$, $i_A : A \rightarrow M$ and $i_B : B \rightarrow M$, such that

$$p_A \circ i_A = 1_A, p_B \circ i_B = 1_B, p_A \circ i_B = 0, p_B \circ i_A = 0,$$

and

$$i_A \circ p_A + i_B \circ p_B = 1_M.$$

Represented by the following diagram:



Definition 1.44

(Richter (2020) Definition 7.2.1)

A category $\mathcal{A}$ is called additive if it is a preadditive category and it has binary biproduct.

Definition 1.45

Let $\mathcal{A}, \mathcal{B}$ be additive categories. A functor $F : \mathcal{A} \rightarrow \mathcal{B}$ is called additive if it preserves finite biproducts.

Definition 1.46

A ringoid is a small preadditive category $\mathcal{A}$, also called ring with many objects when there is more than one object. A left $\mathcal{A}$ -module is an additive functor $F : \mathcal{A} \rightarrow \mathbf{Ab}$, analogously a right $\mathcal{A}$ -module is an additive functor $F : \mathcal{A}^{op} \rightarrow \mathbf{Ab}$.

Remark 1.47

The word ringoid is used here because a ring can be identified as a small preadditive category with one object, similar with the previous example of identifying a monoid as a category with one object in Example 1.3. Concretely, A small preadditive category $\mathcal{R}$ with exactly one object $*$ can be translated as a ring R by considering $Hom(*, *)$ as the additive group R , and the composition as multiplication. The association about left and right module follows similarly.

Definition 1.48

An abelian category $\mathcal{A}$ is a preadditive category that satisfies the following:

- There exists a zero object in $\mathcal{A}$.
- The category $\mathcal{A}$ has all finite biproducts.
- Every morphism has a kernel and a cokernel.
- Every monomorphism is a kernel, and every epimorphism is a cokernel.

Lemma 1.49

(Richter (2020) Proposition 7.1.11)

Let $\mathcal{A}$ be a preadditive category and $A, B \in ob(\mathcal{A})$. Then the following are equivalent:

- The coproduct of A and B exists in $\mathcal{A}$.
- The product of A and B exists in $\mathcal{A}$.
- The biproduct of A and B exists in $\mathcal{A}$.

Proof.

(Richter (2020) Proposition 7.1.11)

It is obvious that since biproduct of A and B is both product and coproduct, the existence of biproduct guarantee the existence of product and coproduct.

Assume the product of A and B exists. Then, we have projection $p_A : A \times B \rightarrow A$ and $p_B : A \times B \rightarrow B$. Consider $1_A : A \rightarrow A$ and $0 : A \rightarrow B$,

by the universal property of product, we have that there exists a morphism $i_A : A \rightarrow A$ such that $p_A \circ i_A = 1_A$ and $p_B \circ i_A = 0$. This prove that the biproduct of A and B , which is the product of A and B , exists. The case for coproduct follows immediately. $\square$

Lemma 1.50

Let $\mathcal{A}$ be an abelian category and $f, g : A \rightarrow B$ for some $A, B \in \mathcal{A}$. We have the equation:

$$eq(f, g) = ker(f - g).$$

Proof.

Since $ker(f - g) = eq(f - g, 0)$ by definition. Let $k : ker(f - g) \rightarrow A$ be the canonical morphism. By $(f - g) \circ k = 0$, we have that $f \circ k = g \circ k$. By the universal property of kernel, we have that is a equalizer of f, g . Since equalizer is unique, thus proving that $ker(f - g) = eq(f, g)$. $\square$

Theorem 1.51

$R\text{-}\mathbf{Mod}$ is an abelian category for any ring R .

Proof.

We first show that $R\text{-}\mathbf{Mod}$ is a preadditive category. For any two module morphisms with same domain and codomain in $R\text{-}\mathbf{Mod}$, say $f, g : M_1 \rightarrow M_2$, we define the additive structure as follows:

- $(f + g)(m) = f(m) + g(m)$ for all $m \in M_1$.
- $0(m) = 0_{M_2}$, for all $m \in M_1$ where 0 is the zero morphism and 0_{M_2} is the additive identity element in the module M_2 .
- $(-f)(m) = -f(m)$ for all $m \in M_1$.

one can easily check that this form an abelian group for $(R\text{-}\mathbf{Mod}(M_1, M_2), +)$. Hence, $R\text{-}\mathbf{Mod}$ is preadditive.

For finite biproducts, one can see that the binary products in $R\text{-}\mathbf{Mod}$, namely products of two objects in the category, is the direct products of two modules, which again is a R -module and thus exist in the category. By the 1.49, we can see that any finite biproduct exists in the category.

For the existence of zero object, we need to show that the zero module, 0 , is the zero object in this category. For any module $M \in ob(R\text{-}\mathbf{Mod})$, there is a module morphism, which is the constant function that sends each element in M to 0 ; hence, the zero module is terminal. Similarly, since every module contains a zero element, there is an inclusion map from the zero module to any module M that sends 0 to 0_M , which is a module morphism; thus, the zero module is initial. Therefore, the zero module is the zero object

in $R\text{-Mod}$.

For finite biproducts, one can see that the binary products in $R\text{-Mod}$ is the direct products of two modules, which again is a R-module and thus exist in the category. By the Lemma 1.49, we can see that any finite biproduct exists in the category.

For any morphism $\phi \in R\text{-Mod}(M_1, M_2)$, the kernel is $\ker(\phi) = \{m \in M_1 : \phi(m) = 0_{M_2}\}$ with the inclusion map $i : \ker(\phi) \rightarrow M_1$. Dually the cokernel of ϕ is the quotient module $M_2/\phi(M_1)$. Both the kernel and cokernel are R-module themselves for any module homomorphism. Thus every morphism in the category has a kernel and a cokernel.

According to Example 1.3, we know that the monomorphisms in $R\text{-Mod}$ are injective, which implies that $M_1 \cong f(M_1)$. Then for any monomorphism $f : M_1 \rightarrow M_2$, f is the kernel of $\pi : M_2 \rightarrow M_2/f(M_1)$. Dually, for any epimorphism $g : M_3 \rightarrow M_4$, g is surjective, which means that g is the cokernel of $\eta : M_3/g(M_4) \rightarrow M_3$. $\square$

1.6 Exactness

In this section we introduce an critical concepts called exact sequence. It is frequently used in homological algebra and in the study of abelian categories.

Definition 1.52

(MacLane (1998) Ch.VIII section 3. Proposition 1.)

In an abelian category, the image and coimage of f is $\text{im}(f) = \ker(\text{coker}(f))$ and $\text{coim}(f) = \text{coker}(\ker(f))$ respectively.

The following theorem is beyond the scope of this essay. It is introduced solely for the purpose of self containment.

Theorem 1.53

(MacLane (1998) Ch.VIII section 3. Proposition 1.)

In an abelian category, for every morphism f , there exists a monomorphism $\text{im}(f)$ and epimorphism $\text{coim}(f)$ such that $f = \text{im}(f) \circ \text{coim}(f)$.

Definition 1.54

An exact sequence in an abelian category $\mathcal{A}$ is a sequence of morphisms in $\{f_1, \dots, f_n\}$ such that it satisfies $\text{im}(f_i) = \ker(f_{i+1})$.

Among exact sequences, short exact sequences, which will be defined below, are of particular importance in category theory and theory of modules. It is frequently used in throughout this essay.

Definition 1.55

A short exact sequence in an abelian category is an exact sequence of the form

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0.$$

Remark 1.56

Notice that if we have a short exact sequence

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0.$$

Since $\ker(f) = \{0\}$ and $\operatorname{im}(g) = C$, therefore we have f is monomorphism and g is epimorphism.

Lemma 1.57

Let $A, B, C \in \mathbf{Ab}$ with $B \subseteq A \subseteq C$. The following sequence is exact.

$$0 \rightarrow A/B \xrightarrow{f} C/B \xrightarrow{g} C/A \rightarrow 0.$$

Where f is the inclusion map and g is the identity map restricted on C/A .

Proof.

Since $A \subseteq C$, the inclusion map $f : A/B \rightarrow C/B$ is injective, which means that $\ker(f) = \{0\} = \operatorname{im}(0)$. Since $B \subseteq A$, the map induced by identity $g : C/B \rightarrow C/A$ defined as $c + B \mapsto c + A$ is surjective. Thus showing that $\operatorname{im}(g) = C/A = \ker(0)$.

It remains to show that $\operatorname{im}(f) = \ker(g)$, which it follows immediately that any element in $\operatorname{im}(f)$ is of the form of $a + B$ for $a \in A$, and $a + B$ is zero in C/A . $\square$

Definition 1.58

(Riehl (2014) Proposition 4.5.10)

Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

be a short exact sequence in $\mathcal{A}$. A functor $F : \mathcal{A} \rightarrow \mathcal{B}$ is called left exact if

$$0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$$

is an exact sequence in $\mathcal{B}$. Respectively, F is called right exact if

$$F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow 0$$

is an exact sequence in $\mathcal{B}$.

1.7 Presentable Categories

In this section, the main focus is on the proof of $R\text{-}\mathbf{Mod}$ being a finitely presentable category, which will be important in the next chapter.

Definition 1.59

(Adamek and Rosicky (1994) p.8)

A non-empty partially ordered set is called directed (or upward directed) if each pair of elements has an upper bound in the set.

Definition 1.60

A directed colimit (also called direct limit) in $\mathcal{C}$ is a colimit of functor $F : (I, \leq) \rightarrow \mathcal{C}$ where $(I, \leq)$ denotes the category corresponding to the directed set $(I, \leq)$ (the correspondence is mentioned in Example 1.3). The functor F is called a directed diagram and the collection $(F(i), f_{ij} = F(i \rightarrow j))_{i,j \in I}$, where $i \rightarrow j$ denotes the unique morphism (there can only have one morphism) from i to j , is called a directed system.

Remark 1.61

Notice that by the definition of category $(I, \leq)$, we have $\exists f : i \rightarrow j$ in the category $\iff i \leq j$ in the directed set. Also, since any pair of elements has an upper bound, any finite subset $J \subseteq I$ also has an upper bound in I .

Definition 1.62

(Adamek and Rosicky (1994) Definition 1.1)

An object $C \in \mathcal{C}$ for some category $\mathcal{C}$ is called finitely presentable if the hom-functor

$$H^C = \mathcal{C}(C, -) : \mathcal{C} \rightarrow \mathbf{Set}$$

preserve directed colimits.

Lemma 1.63

(Adamek and Rosicky (1994) Definition 1.1)

An object $C \in \mathcal{C}$ is finitely presentable if and only if for each directed diagram $F : (I, \leq) \rightarrow \mathcal{C}$, each colimit cocone $(i \xrightarrow{p_i} D)$ and each $f : C \rightarrow D$, there exists $i \in I$ such that

- f factorizes through p_i : $f = p_i \circ g$ for some $g : C \rightarrow i$
- with the factorization being essentially unique in the sense that if there is another morphism $g' : C \rightarrow i$,

$$p_i \circ g' = f = p_i \circ g \implies F(i \rightarrow j) \circ g' = F(i \rightarrow j) \circ g$$

for some $j \geq i$.

The proof of this lemma is beyond the scope of this essay.

Lemma 1.64

(Adamek and Rosicky (1994) Example 1.2)

Let $\mathcal{A}$ be a small category, every hom-functor, $H^A : \mathcal{A} \rightarrow \mathbf{Set}$ for some $A \in ob(\mathcal{A})$, is a finitely presentable object of $[\mathcal{A}, \mathbf{Set}]$.

Proof.

We want to show that H^{H^A} (or rewritten as) $[\mathcal{A}, \mathbf{Set}](H^A, -) : [\mathcal{A}, \mathbf{Set}] \rightarrow \mathbf{Set}$ preserves directed colimit. Explicitly, we want to prove the following,

$$\varinjlim_{i \in I} [\mathcal{A}, \mathbf{Set}](H^A, X_i) = [\mathcal{A}, \mathbf{Set}](H^A, \varinjlim_{i \in I} X_i),$$

for some directed set $(I, \leq)$ and directed system $(X_i, f_{ij})_{i,j \in I}$.

We will need to use the Yoneda lemma here. Recall that for a locally small category $\mathcal{A}$ (a small category must be locally small), there is a bijection between $[\mathcal{A}, \mathbf{Set}](H^A, X)$ and $X(A)$. Consider the diagram of the directed colimit $\varinjlim_{i \in I} X_i$.

$$\begin{array}{ccc} X_j & \xleftarrow{f_{ij}} & X_i \\ & \searrow p_j \quad \swarrow p_i & \\ & \varinjlim_{i \in I} X_i & \end{array}$$

Notice that we can consider the diagram of the functors applying on an object $A \in ob(\mathcal{A})$.

$$\begin{array}{ccc} X_j(A) & \xleftarrow{f_{ijA}} & X_i(A) \\ & \searrow p_{jA} \quad \swarrow p_{iA} & \\ & \varinjlim_{i \in I} X_i(A) & \end{array}$$

Then we can extend the graph by the bijection given by Yoneda lemma

as follows.

$$\begin{array}{ccccc}
X_j(A) & \xleftarrow{f_{ijA}} & & X_i(A) & \\
& \searrow p_{jA} & & \swarrow p_{iA} & \\
& & \varinjlim_{i \in I} X_i(A) & & \\
\uparrow h_j & & \downarrow & & \uparrow h_i \\
[\mathcal{A}, \mathbf{Set}](H^A, X_j) & \dashrightarrow & & [\mathcal{A}, \mathbf{Set}](H^A, X_i) & \\
& \searrow & \downarrow h \varinjlim_{i \in I} X_i & \swarrow & \\
& & [\mathcal{A}, \mathbf{Set}](H^A, \varinjlim_{i \in I} X_i) & &
\end{array}$$

Where h_i for $i \in I$ are the bijections whose existence given by Yoneda lemma. Notice that $([\mathcal{A}, \mathbf{Set}](H^A, X_i), [\mathcal{A}, \mathbf{Set}](H^A, f_{ijA}))_{i,j \in I}$ forms a directed system. By the translations, it is immediate that $[\mathcal{A}, \mathbf{Set}](H^A, \varinjlim_{i \in I} X_i)$ is indeed the directed colimit of $([\mathcal{A}, \mathbf{Set}](H^A, X_i))_{i \in I}$. Thus proving the equation

$$\varinjlim_{i \in I} [\mathcal{A}, \mathbf{Set}](H^A, X_i) = [\mathcal{A}, \mathbf{Set}](H^A, \varinjlim_{i \in I} X_i).$$

□

Definition 1.65

M is said to be a finitely generated R -module if it is a R -module with a finite subset $\{m_1, \dots, m_n\} \subseteq M$ such that for any $m \in M$, m is a R -linear combination of $\{m_1, \dots, m_n\}$. Explicitly, $m = r_1 m_1 + \dots + r_n m_n$ for some $r_i \in R$.

Definition 1.66

M is said to be a finitely presentable R -module if it is a R -module that satisfies the exact sequence described below.

$$R^m \rightarrow R^n \rightarrow M \rightarrow 0$$

0 is the zero module.

Remark 1.67

The definition given above is equivalent as saying that M is finitely generated and the set of relations that a certain set of generators is subject to is also finitely generated. Let $R^m \xrightarrow{\phi} R^n \xrightarrow{\psi} M \xrightarrow{0} 0$ be the exact sequence. Due to the property of the exact sequence, we have that ψ is surjective.

Consider the standard generator $e_i = (0, \dots, 1, \dots, 0)$ of R^n and their image under ψ . Notice that this image completely determines ψ because it is a module homomorphism. Then we can conclude that M is generated by $m_i = \psi(e_i)$ for $1 \leq i \leq n$. Explicitly, for any $m \in M$, $m = \psi(\vec{r})$ for some $\vec{r} \in R^n$ by surjectivity, and $\vec{r} = \sum_i a_i e_i$ for some $a_i \in R$ with $i = 1, \dots, n$. Thus we have the following equation:

$$m = \psi(\vec{r}) = \psi\left(\sum_i a_i e_i\right) = \sum_i a_i \psi(e_i) = \sum_i a_i m_i$$

This only provided us finitely generated module, consider the morphism $R^m \xrightarrow{\phi} R^n$. Notice that $\ker(\psi) = \text{im}(\phi)$, consider $\psi(\sum_i c_i e_i) = \sum_i c_i m_i = 0$ for some $c_i \in R$. Since $\sum_i c_i e_i \in \ker(\psi) = \text{im}(\phi)$, there is some $\vec{b} \in R^m$ such that $\phi(\vec{b}) = \sum_i c_i e_i$.

We call any equations of the form $\sum_i d_i m_i = 0$ for $d_i \in R$, a relation that the set of generators m_i is subject to. By considering the standard generator of R^m we can see that the set of relations that m_i are subject to is finitely generated. This give finitely presentability.

Theorem 1.68

A finitely presentable module is a finitely presentable object in the category $R\text{-Mod}$.

Proof.

To prove this theorem, we want to check that any morphism from a finitely presentable module to a directed colimit will have an essentially unique factorization through the cocone. In order to do that, we will want to write out the explicit formula of the directed colimit in the category of modules.

Claim: Let $F : (I, \leq) \rightarrow R\text{-Mod}$ be a directed diagram, $M_i = F(i)$ and $f_{ij} = F(i \rightarrow j)$ for any $i, j \in (I, \leq)$. We claim that the directed colimit of this diagram $\varinjlim_{i \in I} M_i$ is of the following form.

$$\varinjlim_{i \in I} M_i = \{(x, i) : x \in M_i, i \in I\} / \sim$$

Where $\sim$ is an equivalence relation defined as:

$$(x, i) \sim (y, j) \iff \exists k \geq i, j \text{ such that } f_{ik}(x) = f_{jk}(y).$$

The addition of the module is defined as

$$[(x, i)] + [(y, j)] := [(f_{ik}(x) + f_{jk}(y), k)]$$

with $k \geq i, j$.

Since every $p_i : M_i \rightarrow \varinjlim M_i$ sends 0_{M_i} to $0_{\varinjlim M_i}$, we have that $0_{\varinjlim M_i} = [(0_{M_\eta}, \eta)]$ for some η . Notice that $f_{ij} : M_i \rightarrow M_j$ sends 0_{M_i} to 0_{M_j} , therefore $(0_{M_i}, i) \sim (0_{M_j}, j)$ for any $i, j \in I$.

The additive inverse of $[(x, i)]$ is simply defined as $[(-x, i)]$ since $x \in M_i$ have an additive inverse $-x \in M_i$.

The equivalence relations can be translated as “two elements of two modules are equal if there is a greater (in the sense of the partial ordering of the directed set) module in which the image under the canonical morphism (the morphism induced by the directed set) of the two elements is the same.” One can check that the set is indeed a R -module under this definition.

Proof of the claim:

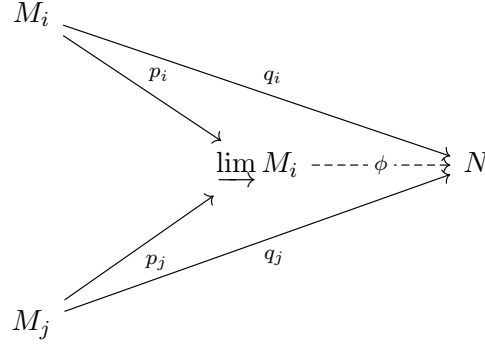
Consider the following diagram,

$$\begin{array}{ccc} M_i & & \\ \downarrow f_{ij} & \searrow p_i & \\ & & \varinjlim M_i \\ & \nearrow p_j & \\ M_j & & \end{array}$$

Where $p_l : M_l \rightarrow \varinjlim M_i$ is defined as $p_l(x) = [(x, l)]$ for any $l \in I$.

Consider $m \in M_i$ and let $m' = f_{ij}(m)$, then $p_i(m) = [(m, i)]$ and $p_j \circ f_{ij}(m) = p_j(m') = [(m', j)]$. According to the definition of the equivalence relation, we can conclude that $(m, i) \sim (m', j)$ by $1_{M_j}(m') = m' = f_{ij}(m)$. Thus, the diagram commutes.

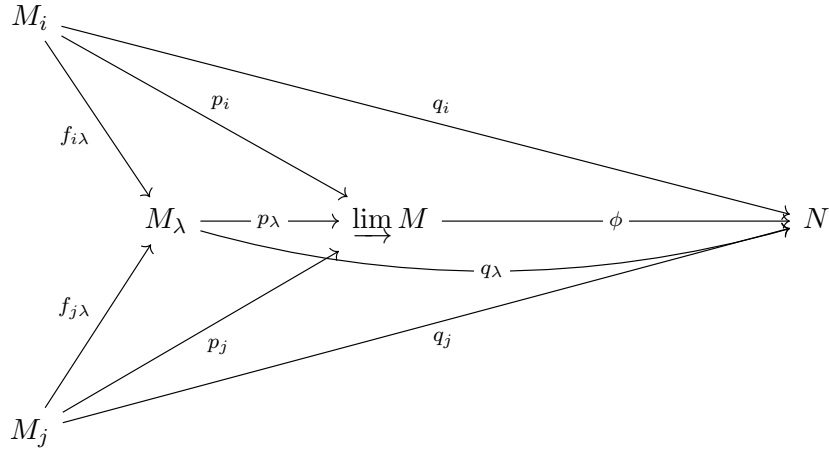
Furthermore, consider any cocone $(M_i \xrightarrow{q_i} N)$,



We can define a morphism $\phi : \varinjlim M_i \rightarrow N$ by $\phi([(x, i)]) := q_i(x)$ for any equivalence classes $[(x, i)] \in \varinjlim M_i$. Since by definition $p_i(x) = [(x, i)]$, the commutativity of diagram above follows immediately.

To check if ϕ is a well defined morphism, let $(x, i) \sim (y, j)$ for some $x \in M_i$ and $y \in M_j$, we want to show that $\phi([(x, i)]) = \phi([(y, j)])$, namely choosing different representative of the same equivalence classes would not affect the output of ϕ .

Let $\lambda \geq i, j$ be such that $f_{i\lambda}(x) = f_{j\lambda}(y)$, then we have the commutative diagram as follows.



This diagram commutes because $(M_i \xrightarrow{q_i} N)$ is a cocone by assumption and $(M_i \xrightarrow{q_i} \varinjlim M)$ is also a cocone by definition.

Here is the argument for ϕ being well defined.

$$\begin{aligned}
\phi([(x, i)]) &= q_i(x) \\
&= q_\lambda \circ f_{i\lambda}(x) \\
&= q_\lambda \circ f_{j\lambda}(y) \\
&= q_j(y) \\
&= \phi([(y, j)])
\end{aligned}$$

■

Now, let M be a finitely presentable module with generators $\{m_1, \dots, m_n\}$, relations generated by

$$\begin{cases} a_{11}m_1 + \dots + a_{1n}m_n = 0 \\ \dots \\ a_{t1}m_1 + \dots + a_{tn}m_n = 0 \end{cases}$$

and $f : M \rightarrow \varinjlim M_i$. We need to show that there exists a $\tilde{f} : M \rightarrow M_q$ for some $q \in I$ such that the following diagram commutes:

$$\begin{array}{ccc}
& & M \\
& \nwarrow \tilde{f} & \downarrow f \\
M_q & & \varinjlim M_i \\
& \searrow p_q & \\
& &
\end{array}$$

There is an algorithm on defining $\tilde{f}$:

- Fix a set of generators $\{m_1, \dots, m_n\}$ and a set of relation generators $\{(a_{11}, \dots, a_{1n}), \dots, (a_{t1}, \dots, a_{tn})\}$ for M .
- Choose a set representatives of the equivalence classes $[(x_j, i_j)] = f(m_j)$ for each $1 \leq j \leq n$. Let the set be $\{(x_1, i_1), \dots, (x_n, i_n)\}$.
- Since $(I, \leq)$ is a directed set, we can find a $k \in I$ such that $k \geq i_1, \dots, i_n$. Then we can consider the set $\{(f_{i_1k}(x_1), k), \dots, (f_{i_nk}(x_n), k)\}$ where f_{i_jk} is the morphism from M_{i_j} to M_k .
- Notice that for any $m = b_1m_1 + \dots + b_nm_n$,

$$\begin{aligned}
f(m) &= f(b_1m_1) + \dots + f(b_nm_n) \\
&= b_1[(x_1, i_1)] + \dots + b_n[(x_n, i_n)] \\
&= b_1[(f_{i_1k}(x_1), k)] + \dots + b_n[(f_{i_nk}(x_n), k)] \\
&= [(b_1f_{i_1k}(x_1) + \dots + b_nf_{i_nk}(x_n), k)] \\
&= p_k(b_1f_{i_1k}(x_1) + \dots + b_nf_{i_nk}(x_n)).
\end{aligned}$$

For each relation generator $(a_{h1}, \dots, a_{hn})$. Observe that

$$p_k(a_{h1}f_{i_1k}(x_1) + \dots + a_{hn}f_{i_nk}(x_n)) = f(0) = [(0, \zeta)],$$

for some M_ζ . Let λ_h be the upper bound of ζ and k such that

$$a_{h1}(f_{k\lambda_h} \circ f_{i_1k}(x_1)) + a_{h2}(f_{k\lambda_h} \circ f_{i_2k}(x_2)) + \dots + a_{hn}(f_{k\lambda_h} \circ f_{i_nk}(x_n)) = 0.$$

- Let λ be the upper bound of $\lambda_1, \dots, \lambda_t$.
- We defined $\tilde{f} : M \rightarrow M_\lambda$ by $\tilde{f}(m_j) := f_{i_j\lambda}(x_j)$ for each $1 \leq j \leq n$.

Notice that $\tilde{f}$ is a module homomorphism by definition, for any $m \in M$, if $m = c_1m_1 + \dots + c_nm_n$, then $\tilde{f}(m) = c_1\tilde{f}(m_1) + \dots + c_n\tilde{f}(m_n)$.

Consider $m = c_1m_1 + \dots + c_nm_n = d_1m_1 + \dots + d_nm_n$, then we have

$$0 = (c_1 - d_1)m_1 + \dots + (c_n - d_n)m_n,$$

which is a relation in M . Since relations of M is finitely generated by the set of relation generators, thus we have $(c_i - d_i) = \mu_{i1}a_{1i} + \dots + \mu_{it}a_{ti}$ for some $\mu_{hk} \in R$ for each i . Notice that,

$$\begin{aligned} \tilde{f}(0) &= (c_1 - d_1)\tilde{f}(m_1) + \dots + (c_n - d_n)\tilde{f}(m_n) \\ &= (\mu_{11}a_{11} + \dots + \mu_{1t}a_{t1})\tilde{f}(m_1) + \dots + (\mu_{n1}a_{1n} + \dots + \mu_{nt}a_{tn})\tilde{f}(m_n) \\ &= (\mu_{11}a_{11}\tilde{f}(m_1) + \dots + \mu_{n1}a_{1n}\tilde{f}(m_n)) + \dots \\ &\quad + (\mu_{t1}a_{1t}\tilde{f}(m_1) + \dots + \mu_{nt}a_{tn}\tilde{f}(m_n)) \\ &= 0 + \dots + 0 = 0 \end{aligned}$$

The last line is given by our definition of λ_h , explicitly since $\tilde{f}(m_j) = f_{i_j\lambda}(x_j) = f_{\lambda_h\lambda} \circ f_{i_j\lambda_h}(x_j)$, and $(\mu_{k1}a_{1k}f_{i_j\lambda_h}(m_1) + \dots + \mu_{nk}a_{kn}f_{i_j\lambda_h}(m_n)) = 0$ by definition. Thus we have that $\tilde{f}(0) = (c_1 - d_1)\tilde{f}(m_1) + \dots + (c_n - d_n)\tilde{f}(m_n) = 0$ implying that $\tilde{f}(m) = c_1\tilde{f}(m_1) + \dots + c_n\tilde{f}(m_n) = d_1\tilde{f}(m_1) + \dots + d_n\tilde{f}(m_n)$. This argument prove that $\tilde{f}$ is well defined for all $m \in M$.

This definition also makes the diagram above commutes. it is determined by the image of the generators of M .

$$\begin{aligned} p_\lambda \circ \tilde{f}(m_j) &= p_\lambda \circ f_{i_j\lambda}(m_j) \\ &= p_\lambda(x_j) \\ &= [(x_j, i_j)] \\ &= f(m_j) \end{aligned}$$

Since a module homomorphism is completely determined by the image of the generators, and thus we can conclude that $p_\lambda \circ \tilde{f} = f$.

Finally, we need to show that this morphism is essentially unique. Explicitly, if we have another morphism $\tilde{g} : M \rightarrow M_\lambda$ with $p_\lambda \circ \tilde{f} = f = p_\lambda \circ \tilde{g}$, then we need to show $f_{\lambda\theta} \circ \tilde{f} = f_{\lambda\theta} \circ \tilde{g}$ for some $\theta \geq \lambda$.

Consider the following diagram.

$$\begin{array}{ccc}
 & M & \\
 \tilde{f} \swarrow & & \downarrow f \\
 M_\lambda & & \varinjlim M_i \\
 p_\lambda \searrow & & \\
 & &
 \end{array}$$

It follows from the assumption that

$$p_\lambda \circ (\tilde{f} - \tilde{g}) = 0,$$

where 0 denotes the zero morphism here. In other words $p_\lambda \circ (\tilde{f} - \tilde{g})(m) = 0_{\varinjlim M_i}$ for any $m \in M$.

Consider the generators of $\{m_1, \dots, m_n\} \in M$, let $e_i := \tilde{f}(m_i) - \tilde{g}(m_i) \in M_\lambda$. $p_\lambda(e_i) = [(e_i, \lambda)] \sim [(0_{M_{\eta_i}}, \eta_i)]$. This equivalence relations is provided by the definition of zero element in the limit. By the definition of the equivalence relation, we can find a M_{κ_i} such that $f_{\eta_i \kappa_i}(e_i) = f_{\eta_i \kappa_i}(0_{\eta_i}) = 0_{\kappa_i}$ for each e_i . Then, consider the upper bound for $\kappa_1, \dots, \kappa_n$, denoted as κ .

Notice that $\kappa \geq \lambda$ have the property $f_{\lambda \kappa}(e_i) = 0_\kappa$ for all e_i . Hence,

$$f_{\lambda \kappa} \circ (\tilde{f} - \tilde{g}) = 0.$$

Because the generators determine the module morphism, we can conclude that $f_{\lambda \kappa} \circ \tilde{f} = f_{\lambda \kappa} \circ \tilde{g}$. □

Definition 1.69

A category $\mathcal{C}$ is called locally finitely presentable if it is cocomplete and has a set A of finitely presentable objects such that every object in $\mathcal{C}$ is a directed colimit of objects in A .

Theorem 1.70

$R\text{-}\mathbf{Mod}$ is a locally finitely presentable category.

Proof.

We can show that $R\text{-}\mathbf{Mod}$ is small cocomplete by the fact that $R\text{-}\mathbf{Mod}$ is an abelian category and Theorem 1.38. Since an abelian category must contain all coproduct and cokernel by definition, and one can check that the cokernel of kernel of $f - g$ is the coequalizer of f and g , namely $\text{coker}(\ker(f - g)) = \text{coeq}(f, g)$, which is also contained in an abelian category. Then by Theorem 1.38, we have that $R\text{-}\mathbf{Mod}$ is small cocomplete.

Consider any object M in $R\text{-}\mathbf{Mod}$. Fix a set $G = \{m_\lambda : \lambda \in \Lambda\}$ of generators of M , and a set of relations $Rel = \{\vec{r} \in \bigoplus_{\lambda \in \Lambda} R : \vec{r} \cdot \vec{m} = 0\}$ where $\vec{m}$ is a list of elements in G (the existence of such a list is given by well ordering principle). Notice that if Λ is an infinite set, then we can not interchange $\prod$ with $\bigoplus$ in the definition of Rel , for infinite sum is not always defined in any modules.

For any finite subset $G' \subseteq G$, and for any

$$Rel' = \{\vec{r} = (r_\lambda)_{\lambda \in \Lambda} \in Rel : m_\lambda \notin G' \implies r_\lambda = 0\} \subseteq Rel.$$

We can construct a finitely presentable module $M_{G' Rel'}$ that is freely generated by G' subject to relations generated by Rel' . Notice that this definition of Rel' makes sure that any $\vec{r} \in Rel'$ is a relation only on elements of G' .

Define a collection $(M_{G_i Rel_i}, f_{ij})_{i,j \in I}$ for some index set I as follows.

$$\exists f_{ij} : M_{G_i Rel_i} \rightarrow M_{G_j Rel_j} \iff G_i \subseteq G_j \wedge Rel_i \subseteq Rel_j.$$

Notice that the “upper bound” for any pair of modules exists, it is constructed as $M_{G_i \cup G_j Rel_i \cup Rel_j}$. Therefore, this is indeed a directed system, explicitly, we can induced an ordering on I by the above condition, namely $i \leq j \iff \exists f_{ij}$, making $(I, \leq)$ a directed set.

Denote $M_{G_i Rel_i}$ as M_i . Construct a functor $F : (I, \leq) \rightarrow R\text{-}\mathbf{Mod}$ by $F(i) = M_i$, $i_{ij} = F(i \rightarrow j)$. Then denote the directed colimit of diagram F as $\varinjlim M_i$. Note that $M_{G_i Rel_i} \rightarrow M_{G_j Rel_j}$ is an inclusion map for any $M_{G_i Rel_i}, M_{G_j Rel_j}$. As shown in the proof of Theorem 1.68, $\varinjlim_{i \in I} M_i = \{(x, i) : x \in M_i, i \in I\} / \sim$, with the equivalence relations defined as:

$$(m, M_{G_x Rel_x}) \sim (m', M_{G_y Rel_y}) \iff \exists z \geq x, y \text{ such that } i_{xz}(m) = i_{yz}(m').$$

Note that every element $m \in M$, we can find a finite set of generators that generate m (the definition of generators guarantees that every elements is a finite R -linear combination of the generators). Hence we can define

a morphism from the original module M to the directed colimit $\varinjlim M_i$ by $\phi(m) = [(m, M_{G' Rel'})]$ for some G' that generates m and some finite set Rel' .

To check that ϕ is well defined under this definition, we need to consider that if $m = \vec{r} \cdot \vec{m} = \vec{s} \cdot \vec{m}$ for $\vec{m}$ is the list of generators of M , and $\vec{r}, \vec{s}$ are some tuples of element of R . Define $G_a = \{m_\lambda : \lambda \in \Lambda \wedge r_\lambda \neq 0\}$ and $G_b = \{m_\lambda : \lambda \in \Lambda \wedge s_\lambda \neq 0\}$, and define any appropriate Rel_a and Rel_b respectively. We want to show that $(\vec{r} \cdot \vec{m}, M_{G_a Rel_a}) \sim (\vec{s} \cdot \vec{m}, M_{G_b Rel_b})$. Let k be the upper bound of a, b such that $M_k = M_{G_a \cup G_b Rel_a \cup Rel_b}$, then because i_{ak} and i_{bk} are both inclusion maps, we have $i_{ak}(\vec{r} \cdot \vec{m}) = \vec{r} \cdot \vec{m} = m = \vec{s} \cdot \vec{m} = i_{bk}(\vec{s} \cdot \vec{m})$.

By the argument above, we can also conclude that the morphism

$$\phi^{-1}([(m, M_{G_x Rel_x})]) = m$$

is well-defined. Since $(m, M_{G_x Rel_x}) \sim (m', M_{G_y Rel_y})$ implies that $m = i_{xz}(m) = i_{yz}(m') = m'$. As one can see that it is the inverse of the morphism ϕ defined above, and thus we have shown that $M \cong \varinjlim M_i$. $\square$

2 PP condition in the Language for right R-modules

2.1 Model theory of modules

In this section, we will briefly introduce the basic definitions of language, terms and formulas in model theory, in order to formalize the setting in our study of pp-conditions.

Definition 2.1

(Kirby (2019) Definition 1.1)

A language L is a collection of the following:

- A set of relation symbols R .
- A set of function symbols F .
- A set of constant symbols C .
- A positive natural number called arity assign for each relation and function symbol.
- A set of logical symbols: $\{\rightarrow, \neg, \wedge, \vee, \forall, \exists, =, v_0, v_1, \dots\}$, where $v_0, v_1, \dots$ are called variable symbols.

The language is denoted as $L = (R, F, C)$.

Languages themselves are completely formal and carry no intrinsic meaning. It determines what can be expressed.

Definition 2.2

(Kirby (2019) Definition 1.2)

An L -structure A is collection of the following:

- A set A called the domain of A .
- For each relation symbol R of language L with arity n , a subset of $R^A \subseteq A^n$.
- For each function symbol f of language L with arity n , a function $f^A : A^n \rightarrow A$.
- For each constant symbol c of language L , an element $c^A \in A$.

The L -structure is denoted as $A = (A; R^A, F^A, C^A)$.

A structure provides semantic meaning to the symbols of the language, which is called interpretation. It provides a background to examine all the “words” of the language created by combinations of symbols.

Example 2.3

- The empty language is where $R = F = C = \emptyset$, which is also called the basic language, denoted as $L_\emptyset$.

- (Kirby (2019) Example 1.4)
The language of partially ordered set, $L_{\leq} = (\leq)$, contains only a binary relation symbol $\leq$. An structure of $L_{\leq}$ with domain M is interpreted as a poset where $\leq$ is the subset $\{(a, b) \in M : a \leq b\}$.
- (Kirby (2019) Example 1.5)
The language of groups is $L_{\mathbf{grp}} = (\cdot, (-)^{-1}, 1)$, where $\cdot$ is a binary function symbol, $(-)^{-1}$ is a unary function symbol and 1 is a constant symbol. A group G can be viewed as an $L_{\mathbf{grp}}$ -structure with domain G with binary operation $\cdot^G$, identity element 1^G and an unary function $((-)^{-1})^G$ sending every element $a \in G$ to its inverse, namely $a \mapsto a^{-1}$.
- The language of right R -modules is $L_{\mathbf{Mod-R}} = (0, +, (*_r)_{r \in R})$, where 0 is a constant symbol, $+$ is a binary function symbol and $(*_r)$ is a unary function symbol for each $r \in R$. An right R -module M can be viewed as an $L_{\mathbf{Mod-R}}$ structure with domain M , where 0^M is the additive identity and form an abelian group together with $+^M$, $(*_r)$ interpret as right multiplication by r .

Definition 2.4

(Kirby (2019) Definition 2.1)

For a language L , the set of terms of L is defined recursively as follows.

- Every variables is a term.
- Every constant symbols of L is a term.
- For $t_1, \dots, t_n$ are terms of L and f is a function symbol of L with arity n , $f(t_1, \dots, t_n)$ is a term.
- Only strings of symbols built from the above three clauses in finitely many steps are terms.

Example 2.5

Consider the language of groups $L_{\mathbf{grp}}$, x_i are the variable symbols for $i \in \mathbb{N}$. Then x_1 is a term, similarly the constant symbol 1 is also a term. It can be shown by construction that $1 \cdot x$, 1^{-1} , $((1 \cdot 1)^{-1} \cdot x) \cdot x^{-1}$ are terms.

Definition 2.6

(Kirby (2019) Definition 3.1)

For a language L , the set of formulas of L defined recursively as follows.

- If t_1, t_2 are terms of L , then $t_1 = t_2$ is a formula of L .
- If $t_1, \dots, t_n$ are terms, R is a relation symbol of L with arity n , then $R(t_1, \dots, t_n)$.
- If ϕ, ψ are formulas of L , then $\neg\phi$ and $(\phi \wedge \psi)$.

- If ϕ is a formula and x a variable, then $\exists x(\phi)$ is a formula.
- Only strings of symbols built from the above four clauses in finitely many steps are formulas.

Example 2.7

Consider the language of partially ordered set $L_{\leq}$ with variable symbols x_i for $i \in \mathbb{N}$. Then $x_1 = x_1$ is a formula, similarly $x_1 \leq x_2$ is also a formula. It can be shown by construction that $\neg(x_1 = x_2)$, $\neg(x_1 = x_2) \wedge (x_1 \leq x_2)$, $\exists x_3(x_1 \leq x_3)$ are formulas.

2.2 pp-condition

In this section, the main goal is to introduce the pp-condition, which will be defined later, and to prove the Theorem 2.51. This can provide a new perspective on the model theory of modules. The following paragraph will follow Prest (2009), and therefore we will mainly use the category of right module $\mathbf{Mod}\text{-}R$. Despite the fact that the category of left and right modules can be quite different, for a particular ring, all the results proven in chapter 1 about $R\text{-}\mathbf{Mod}$ can be translated directly to $\mathbf{Mod}\text{-}R$ as $\mathbf{Mod}\text{-}R \cong R^{op}\text{-}\mathbf{Mod}$. More generally, most of the results given in the following section can be generalised to many-objects ring, where the generalisation is shown in Prest (2009).

Definition 2.8

(Prest (2009) p.17-18)

Consider a finite homogeneous system of R -linear equations in right modules:

$$\bigwedge_{j=1}^m \sum_{i=1}^n x_i r_{ij} = 0,$$

it can also be interpreted as

$$\vec{x}H = 0,$$

where H is the $n \times m$ matrix $(r_{ij})_{ij}$ and $\vec{x} = (x_1, \dots, x_n)$.

It can be regarded as a single condition $\theta(\vec{x})$ on the tuple. The solution set is

$$\theta(M) = \{\vec{a} = (a_1, \dots, a_n) \in M^n : \bigwedge_{j=1}^m \sum_{i=1}^n x_i r_{ij} = 0\}.$$

A pp-condition ϕ is the condition of the form:

$$\exists x_{k+1}, \dots, x_n \bigwedge_{j=1}^m \sum_{i=1}^n x_i r_{ij} = 0$$

or abbreviated as $\exists \vec{x'} \theta(\vec{x}, \vec{x'})$ where $\vec{x} = (x_1, \dots, x_k)$ and $\vec{x'} = (x_{k+1}, \dots, x_n)$.

A subgroup of M^k is called pp-definable in M if the subgroup is of the form:

$$\phi(M) = \{\vec{a} = (a_1, \dots, a_k) \in M^k : \exists a_{k+1}, \dots, a_n \in M \bigwedge_{j=1}^m \sum_{i=1}^n x_i r_{ij} = 0\}.$$

Remark 2.9

From a model theory perspective, a pp-condition is a formula of language of *mod* r build only by $=$, $\wedge$, $(*_r)r \in R$ and $\exists$. In other words, any formula without $\neg$. This type of formulas are of special importance, as shown in the paper by Baur, every first-order formula in the language of R -modules, up to certain restriction, is equivalents to a boolean combination of positive primitive formulas.

Remark 2.10

(Prest (2009) p.19)

Writing H as a block matrix $H = \begin{pmatrix} A \\ -B \end{pmatrix}$. The pp-condition can be rewritten as $\exists \vec{y} (\vec{x}A = \vec{y}B)$, which may be denoted as $B \mid \vec{x}A$ (" B divides $\vec{x}A$ ").

Example 2.11

- Notice that $x = 0$ and $x = x$ are both pp-condition.
- One of the basic examples is the annihilator of r in M . If we consider m and n to be just 1, then the system of equations will be $xr = 0$ where the solution set is the annihilator of r in M .
- Consider $\mathbb{Z}/3\mathbb{Z}$ as a $\mathbb{Z}$ -module. Then,

$$\phi(x) : \exists b, c \in \mathbb{Z}/3\mathbb{Z} \quad (4x + b + 2c = 0) \wedge (3x + 1c = 0) \wedge (2x + 5b + 2c = 0),$$

is a pp-condition. The matrix representing this pp-condition is

$$H_\phi = \begin{pmatrix} 4 & 3 & 2 \\ 1 & 0 & 5 \\ 2 & 1 & 2 \end{pmatrix}.$$

Rewriting this condition as divisibility condition will give us

$$\exists (b, c) \quad (x) \begin{pmatrix} 4 & 3 & 2 \end{pmatrix} = (b, c) \begin{pmatrix} 1 & 0 & 5 \\ 2 & 1 & 2 \end{pmatrix}.$$

One can check that $(0) \in \phi(\mathbb{Z}/3\mathbb{Z})$ as $(0, 0, 0)H_\phi = 0$. Also $(1) \in \phi(\mathbb{Z}/3\mathbb{Z})$ since $(1, 2, 0)H_\phi = 0$.

- Consider $\mathbb{R}[x]/\langle x^2 \rangle$ as a $\mathbb{R}$ -module where $\langle x^2 \rangle$ is the ideal generated by x^2 . Let $\psi(\vec{x})$ be the pp-condition that is,

$$\psi((a, b)) : \exists c \in \mathbb{R}[x] \quad (a, b, c) \begin{pmatrix} 2x & x^2 & 2x \\ x & 0 & 4x \\ 3x & 1 & 3 \end{pmatrix} = 0.$$

One can check that $(x, 2x) \in \psi(\mathbb{R}[x]/\langle x^2 \rangle)$, since

$$(x, 2x, \frac{-2x}{3}) \begin{pmatrix} 2x & x^2 & 2x \\ x & 0 & 4x \\ 3x & 1 & 3 \end{pmatrix} = 0.$$

Lemma 2.12

(Prest (2009) Lemma 1.1.7)

If $f : M \rightarrow N$ is a morphism and ϕ is a pp-condition then $f(\phi(M)) \leq \phi(N)$.

Proof.

(Prest (2009) Lemma 1.1.7)

Let $\vec{a} \in \phi(M)$, then there exists $\vec{b}$ such that $(\vec{a}, \vec{b}) \in \theta(M)$. Then $(f(\vec{a}), f(\vec{b})) \in \theta(N)$ by the preservation of addition and scalar multiplication of module morphism, thus $f(\vec{a}) \in \phi(N)$. $\square$

Definition 2.13

(Prest (2009) p.23)

Let ϕ be a pp-condition, then the pp-functor $F_\phi : \mathbf{Mod}\text{-}R \rightarrow \mathbf{Ab}$ is defined as follows:

1. $M \mapsto \phi(M)$.
2. $f \mapsto F_\phi(f)$ for any $f : M \rightarrow N$ where $F_\phi(f) : \phi(M) \rightarrow \phi(N)$ is defined as follows.

For any $\vec{a} = (a_1, \dots, a_k) \in \phi(M)$, $F_\phi(f)(\vec{a}) = (f(a_1), \dots, f(a_k)) \in \phi(N)$.

One can check that this is well defined by the reasoning of Lemma 2.12.

Definition 2.14

(Prest (2009) p.25)

Let ϕ, ψ be pp-conditions that have the same number of free variables. If they satisfy $\psi(M) \leq \phi(M)$ in every module M , we say that ψ implies ϕ or $\psi \rightarrow \phi$ or $\psi \leq \phi$.

These pp-conditions are said to be equivalent, denoted as $\psi \cong \phi$, if $\phi \leq \psi$ and $\psi \leq \phi$.

A pp-pair ϕ/ψ is a pair of pp-conditions such that $\phi \geq \psi$.

Notice that $\psi(M) \leq \phi(M)$ means that if $\vec{a} \in \psi(M)$ then $\vec{a} \in \phi(M)$. In other words, $\vec{a}$ satisfies the pp-condition ψ in M implies that $\vec{a}$ satisfies ϕ in M . This is the reason for our notion of implication.

Remark 2.15

(Prest (2009) p.25)

Recall from Remark 2.10 that any pp-condition with free variables $\vec{x}$ can be written in the form $\exists \vec{y}(\vec{y}B = \vec{x}A)$, which we will denote as $B \mid \vec{x}A$. The following implications are immediate:

- $B \mid \vec{x}A \implies BC \mid \vec{x}AC$ for any matrix C ;
- $B \mid \vec{x}A \implies B_0 \mid \vec{x}A$ if $B = B_1B_0$ for some matrices B_1, B_0 ;
- $B \mid \vec{x}A \implies B \mid \vec{x}D$ if $A = A_0B + D$ for some matrices A_0, D .

Lemma 2.16

(Prest (2009) Lemma 1.1.13)

Let $\phi(x)$ be the pp-condition $\exists \vec{y}(\vec{x}, \vec{y})H_\phi = 0$, and $\psi(x)$ be the pp-condition $\exists \vec{z}(\vec{x}, \vec{z})H_\psi = 0$. Then $\psi \leq \phi$ if and only if there are matrices $G = \begin{pmatrix} G' \\ G'' \end{pmatrix}$ and K such that

$$\begin{pmatrix} I & G' \\ 0 & G'' \end{pmatrix} H_\phi = H_\psi K,$$

where I is the $n \times n$ identity matrix and 0 is the $n \times n$ zero matrix with n being the length of $\vec{x}$.

Proof.

(Prest (2009) Lemma 1.1.13)

Suppose that $H_\psi = \begin{pmatrix} A' \\ -B' \end{pmatrix}$ and $H_\phi = \begin{pmatrix} A \\ B \end{pmatrix}$, with A', B' are $n \times m$ and $l \times m$ matrices respectively.

($\implies$)

Assume $\psi \leq \phi$. Consider the morphism from R^m to R^{n+l} given by $\vec{z} \mapsto \begin{pmatrix} A'\vec{z} \\ -B'\vec{z} \end{pmatrix}$. Denote M as the cokernel of this morphism. We can see that M is finitely presented by the following exact sequence:

$$R^m \rightarrow R^{n+l} \rightarrow M \rightarrow 0.$$

Let e_i be the i -th unit element in R^{n+l} . Let denote the image of e_i under the canonical cokernel morphism by a_i if $i \leq n$ and by b_i if $n+1 \leq i \leq n+l$.

By the definition of cokernel and module homomorphism, we can see that $(\vec{a}, \vec{b})$ generates the module M and subject to the relations $(\vec{a}, \vec{b}) \begin{pmatrix} A' \\ -B' \end{pmatrix} = 0$.

Hence $\vec{a} \in \psi(M)$ and therefore $\vec{a} \in \phi(M)$ by assumption. So there is $\vec{d}$ in M such that $(\vec{a}, \vec{d}) \begin{pmatrix} A \\ -B \end{pmatrix} = 0$.

Since $\vec{d}$ is from M and $(\vec{a}, \vec{b})$ generates M , there are matrices G', G'' such that $\vec{d} = (\vec{a}, \vec{b}) \begin{pmatrix} G' \\ G'' \end{pmatrix}$. We have that $(\vec{a}, \vec{b}) \begin{pmatrix} I & G' \\ 0 & G'' \end{pmatrix} \begin{pmatrix} A \\ -B \end{pmatrix} = 0$. Then by finite presentability, we have $\begin{pmatrix} I & G' \\ 0 & G'' \end{pmatrix} \begin{pmatrix} A \\ -B \end{pmatrix} = \begin{pmatrix} A' \\ -B' \end{pmatrix} K$ for some matrix K as the relations of generators $(\vec{a}, \vec{b})$ is finitely generated. The equation prove the implication.

($\Leftarrow$)

Suppose that $\begin{pmatrix} I & G' \\ 0 & G'' \end{pmatrix} H_\phi = H_\psi K$, so $A - G'B = A'K$ and $-G''B = -B'K$.

Then by the implications mentioned in the Remark 2.15, we can show that

$$\begin{pmatrix} A' \\ -B' \end{pmatrix} \Rightarrow \begin{pmatrix} A'K \\ -B'K \end{pmatrix} = \begin{pmatrix} A - G'B \\ -G''B \end{pmatrix} \Rightarrow \begin{pmatrix} A - G'B \\ -B \end{pmatrix} \Rightarrow \begin{pmatrix} A \\ -B \end{pmatrix},$$

as required. $\square$

Definition 2.17

A lattice is a partially ordered set $(L, \leq)$ such that for every pair of elements $a, b \in L$, there exists a least upper bound and a greatest lower bound. We call the least upper bound the “join” and the greatest lower bound the “meet”.

Definition 2.18

(Prest (2009) p.28) Let $\phi(\vec{x}) : \exists \vec{y} \theta(\vec{x}, \vec{y})$ and $\psi(\vec{x}) : \exists \vec{z} \eta(\vec{x}, \vec{z})$ be two pp-conditions in the same free variables. Let $\delta(\vec{x}, \vec{y}, \vec{z})$ be the system of equations which is the union of $\theta(\vec{x}, \vec{y})$ and $\eta(\vec{x}, \vec{y})$ (rename the variables so that $\vec{y}$ and $\vec{z}$ is disjoint). Then we write $\phi \wedge \psi$ as the condition

$$\exists \vec{y}, \vec{z} \delta(\vec{x}, \vec{y}, \vec{z}).$$

Similarly, we denotes the condition

$$\exists \vec{v}, \vec{v}' (\phi(\vec{v}) \wedge \psi(\vec{v}') \wedge (\vec{x} = \vec{v} + \vec{v}')),$$

(where $\vec{z}, \vec{z}'$ are the renaming of $\vec{x}$ that is disjoint) as $\phi + \psi$. Notice that the $\phi \wedge \psi$ defines $\phi(M) \cap \psi(M)$ and $\phi + \psi$ defines $\phi(M) + \psi(M)$.

Definition 2.19

(Prest (2009) p.29) The lattice of pp-conditions, denoted as pp_R^n , is the set of pp-conditions in n free variables, with the implication pre-ordering defined as in Definition 2.14. One can easily check that the conjunction $\wedge$ and the sum $+$ defined in Definition 2.18 are the meet and join operations, respectively.

Given a module M , let $pp^n(M)$ be the (modular)lattice of subgroups of M^n that are pp-definable in M .

Example 2.20

(Prest (2009) p.30)

- Let R be any ring and $I = \langle r_1, \dots, r_m \rangle$ be a finitely generated right ideal of R . We can see that $xI = 0$ is a pp-condition by writing it explicitly as $\bigwedge_1^m xr_i = 0$.

More generally, let I' be a finitely generated submodule of R^n , then a condition on tuple $\vec{x}$ defined by $\vec{x}I' = \{ \vec{x} \cdot \vec{r} : \vec{r} \in I' \} = 0$ is a pp-condition as $\vec{x} \cdot \vec{r} = \sum_1^n x_i r'_i$.

- Let R be any ring and $L = \langle r_1, \dots, r_m \rangle$ be a finitely generated left ideal of R . Consider the condition on $x \in M$ defined as

$$L \mid x \iff x \in ML = \left\{ \sum_i a_i l_i : a_i \in M, l_i \in L \right\}$$

. This is a pp-condition because it is equivalent to $\exists y_1, \dots, y_m (x = \sum_1^m y_i s_i)$.

More generally, let $L = \langle s_1, \dots, s_m \rangle$ be a finitely generated left submodule of R^n , then a condition on tuple $\vec{x}$ defined by $x \in ML = \{ \vec{a} \in M^n : \exists b_1, \dots, b_m (\vec{a} = \sum_1^m b_j \vec{s}_j) \}$ is equivalent $\phi(\vec{x}) : \exists y_1, \dots, y_m (\bigwedge_{i=1}^n x_i = \sum_{j=1}^m y_j s_{ij})$ where $\vec{s}_j = (s_{ij})_i$. This is a pp-condition, with $L = \phi(R_R)$ and $\phi(M) = ML = M\phi(R)$.

Note that for right modules, we have annihilation condition by right ideals and divisibility by left ideals.

Instead of selecting a particular pp-condition and examining the sets it defines across different modules. One may adopt a dual perspective, fix an element within a given module and consider the collection of all the pp-conditions that the element satisfies. This collection is referred to as the pp-type of the element.

Definition 2.21

(Prest (2009) p.31) For a module M and a tuple $\vec{a} = (a_1, \dots, a_n) \in M^n$, the pp-type of $\vec{a}$ in M is

$$pp^M(\vec{a}) = \{\phi : \phi \text{ is a pp-condition and } \vec{a} \in \phi(M)\}.$$

Definition 2.22

(Prest (2009) p.31) The pp-type of $\vec{a}$ is said to be finitely generated by $\phi_0 \in pp^M(\vec{a})$, denoted as $pp^M(\vec{a}) = \langle \phi_0 \rangle$, if $pp^M(\vec{a}) = \{\phi \in pp_R^n : \phi \geq \phi_0\}$.

Remark 2.23

(Prest (2009) p.33) Notice that proving $\phi_0(M) \leq \phi(M)$ for all $\phi \in pp^M(\vec{a})$ is not enough for $\langle \phi_0 \rangle = pp^m \vec{a}$. Since proving $\phi_0 \leq \phi$ requires $\phi_0(M') \leq \phi(M')$ for all modules M' . However, later in the essay, Lemma 2.29 will show us that we only need to check the modules that are finitely generated, which makes proving a pp-type being finitely generated much more efficient.

Definition 2.24

(Prest (2009) p.35) A free realization of a pp-condition ϕ is a finitely presented module M and a tuple $\vec{a}$ from M such that $pp^M(\vec{a}) = \langle \phi \rangle$.

Lemma 2.25

(Prest (2009) Lemma 1.2.2) Let $\vec{a} = (\vec{a}_i)_i \in \prod_i M_i^n = (\prod_i M_i)^n = M$ for some n and i in some index set. Then $pp^M(\vec{a}) = \bigcap_i pp^{M_i}(\vec{a}_i)$.

Proof.

Let $\vec{a} \in \phi(M)$ and $\phi(\vec{x})$ be the condition that $\exists \vec{y}(\vec{x}, \vec{y})H = 0$. Then let $\vec{b} = (\vec{b}_i)_i$ from M such that $(\vec{a}, \vec{b})H = 0$. It gives $(\vec{a}_i, \vec{b}_i)H = 0$ for each i . Hence, $\vec{a}_i$ satisfies ϕ in M_i for each i . For the converse, if $\vec{a}_i$ satisfies ϕ with $\vec{b}_i$ for each M_i , then trivially that $\vec{b} = (\vec{b}_i)_i$ will have that $(\vec{a}, \vec{b})$ satisfies ϕ in M . $\square$

Corollary 2.26

(Prest (2009) Lemma 1.2.3) For any modules and pp-conditions, the operations $\phi(-)$ commutes with product and coproduct. Explicitly, $\phi(\bigoplus_i M_i) = \bigoplus_i \phi(M_i)$ and $\phi(\prod_i M_i) = \prod_i \phi(M_i)$.

Proof.

For product, let $\vec{a} = (a_i)_i \in \prod_i M_i = M$, we have that

$$\begin{aligned} \vec{a} \in \phi\left(\prod_i M_i\right) &\iff \phi \in pp^M(\vec{a}) \\ &\iff \vec{a}_i \in \phi(M_i) \quad \forall i \\ &\iff \vec{a} \in \prod_i \phi(M_i) \end{aligned} \tag{2.25}$$

For coproduct which is just direct sum, use the fact that $\bigoplus_i \phi(M_i) \subseteq \prod_i \phi(M_i)$ and $\phi(\bigoplus_i (M_i)) \subseteq \phi(\prod_i (M_i))$ by the inclusion map and Lemma 2.12. The rest follows analogously from the argument. $\square$

Theorem 2.27

(Prest (2009) Proposition 1.2.14) Every pp-condition has a free realization.

Proof.

(Prest (2009) Proposition 1.2.14)

Let ψ be a pp-condition, the proof of Lemma 2.16 produced a free realization of ψ denoted as $(M, \vec{a})$ in the proof. Explicitly, we can write the pp-condition as $\psi(\vec{x}) : \exists \vec{y}(\vec{x}, \vec{y}) \begin{pmatrix} A \\ -B \end{pmatrix} = 0$ for some $n \times m$ matrix A and $l \times m$ matrix B by Remark 2.10. Define M as the cokernel of the morphism from $R^m \rightarrow R^{n+l}$ defined by $\vec{z} \mapsto \begin{pmatrix} A\vec{z} \\ -B\vec{z} \end{pmatrix}$; Define a_i as the image of the standard free generators e_i of R^{n+l} under the morphism for $i \leq n$ and define b_i as the same image for $n+1 \leq i \leq n+l$.

Consider a pp-condition $\phi \in pp^M(\vec{a})$, we write it as $\phi(\vec{x}) : \exists \vec{y}(\vec{x}, \vec{y}) H_\phi = 0$. Then we can see that there is a tuple $\vec{d}$ from M such that $(\vec{a}, \vec{d})H_\phi = 0$. By the finite presentability of M demonstrated in the proof of Lemma 2.16, we can see $\vec{d} = (\vec{a}, \vec{b}) \begin{pmatrix} G'_d \\ G''_d \end{pmatrix}$ for some matrices G'_d, G''_d . Hence $(\vec{a}, \vec{b}) \begin{pmatrix} I & G'_d \\ 0 & G''_d \end{pmatrix} \begin{pmatrix} A \\ -B \end{pmatrix} = 0$ and using the finite presentability again we have that $\begin{pmatrix} I & G'_d \\ 0 & G''_d \end{pmatrix} \begin{pmatrix} A \\ -B \end{pmatrix} = \begin{pmatrix} I & G'_d \\ 0 & G''_d \end{pmatrix} H_\psi = H_\phi K$ for some matrix K . By the Lemma 2.16, we can see that $\psi \leq \phi$ for any $\phi \in pp^M(\vec{a})$, thus $pp^m(\vec{a}) = \langle \psi \rangle$ as desired. $\square$

The free realisation is important in the sense that any other realisation factors uniquely through it, which largely benefits our study on pp-condition. The following two corollaries will be useful in the proof of our final theorem. However the proof is beyond the scope of this essay.

Corollary 2.28

(Prest (2009) Corollary 1.2.17) If $(C, \vec{c})$ is a free realization of the pp-condition ϕ and M is any module, then $\vec{a}$ from M satisfies ϕ if and only if there is a morphism $f : C \rightarrow M$ with $f(\vec{c}) = \vec{a}$. In other words, $\phi(M) = \{ f(\vec{c}) : f \in R\text{-Mod}(C, M) \}$.

Corollary 2.29

(Prest (2009) Corollary 1.2.23) Consider pp-conditions ψ and ϕ , $\psi \leq \phi$ if and only if $\psi(M) \leq \phi(M)$ for every finitely presented modules M .

Other than functor defined by pp-condition in Definition 2.13, a functor define for pp-pair is also an important notion.

Definition 2.30

(Prest (2009) p.110) Let ϕ/ψ be a pp-pair, then we define $F_{\phi/\psi}$ as a functor from $\mathbf{Mod}\text{-}R$ to $\mathbf{Ab}$ by

$$F_{\phi/\psi}(M) = F_\phi(M)/F_\psi(M) = \phi(M)/\psi(M).$$

Given a morphism $f : M \rightarrow N$ in $\mathbf{Mod}\text{-}R$, the morphism

$$F_{\phi/\psi}(f) : \phi(M)/\psi(M) \rightarrow \phi(N)/\psi(N)$$

is induced by the pp functor $F_\phi(f)$ as follows:

$$F_{\phi/\psi}(f)(m + \psi(M)) = F_\phi(f)(m) + \psi(N).$$

Theorem 2.31

(Prest (2009) Proposition 1.2.31) Any pp-functors commute with directed colimits.

Explicitly, for any pp-condition ϕ and module $M = \varinjlim M_\lambda \in R\text{-}\mathbf{Mod}$, we have $\phi(M) = \varinjlim \phi(M_\lambda)$ with the maps between the $\phi(M_\lambda)$ being those induced by the maps between the M_λ .

The proof of this theorem is beyond the scope of this essay.

Lemma 2.32

(Prest (2009) Lemma 1.2.6) If M is a finitely presented module and $\vec{a}$ is a tuple from M then the pp-type of $\vec{a}$ in M is finitely generated.

Proof.

Let $\vec{b}$ be the set of generators of M . Then we can expand any element in $\vec{a}$ as linear combinations of elements in $\vec{b}$, which is equivalent to find some matrix G such that $\vec{a} = \vec{b}G$. Since M is finitely presentable, then $\vec{b}$ is subject to relations that are finitely generated, hence there exist matrix H such that $\vec{b}H = 0$ generates all the relations of the generators.

Claim: Let $\phi(\vec{x})$ be the pp-condition $\exists \vec{y} \begin{pmatrix} I & 0 \\ -G & H \end{pmatrix} \vec{x} = 0$.

$$pp^M(\vec{a}) = \langle \phi \rangle$$

Proof of the claim: It is trivial by the construction that $\phi \in pp^M(\vec{a})$, because

$$(\vec{a}\vec{b}) \begin{pmatrix} I & 0 \\ -G & H \end{pmatrix} = \begin{pmatrix} \vec{a} - \vec{b}G & 0 + \vec{b}H \end{pmatrix} = 0.$$

We want to show that for any $\psi \in pp^M(\vec{a})$, we have $\psi \geq \phi$. Let $\psi \in pp^M(\vec{a})$ be the condition $\exists \vec{y} \vec{x}A = \vec{y}B$. We have $\vec{a}A = \vec{d}B$ for some $\vec{d}$. Consider any finitely presented module M' and $\vec{x} \in \phi(M')$, then we have $\vec{x} = \vec{y}G$ and $\vec{y}H = 0$ for some $\vec{y}$. Notice that since M is finitely presented, we can write $\vec{d} = \vec{b}G_d$ for some matrix G_d .

$$\begin{aligned}
\vec{a}A &= \vec{d}B \\
\vec{b}GA &= \vec{b}G_dB \\
\vec{b}(GA - G_dB) &= 0.
\end{aligned}$$

Since $\vec{b}(GA - G_dB) = 0$ is a relation that $\vec{b}$ is subjected to, then by the finite presentability of M again, we have $(GA - G_dB) = H\Lambda$ for some matrix Λ . Consider $\vec{x} = \vec{y}G$,

$$\begin{aligned}
\vec{x} &= \vec{y}G \\
\vec{x}A &= \vec{y}GA \\
\vec{x}A &= \vec{y}(H\Lambda + G_dB) \\
\vec{x}A &= \vec{y}H\Lambda + \vec{y}G_dB \\
\vec{x}A &= (\vec{y}H)\Lambda + \vec{y}G_dB \\
\vec{x}A &= \vec{0} + \vec{y}G_dB.
\end{aligned}$$

Let $\vec{z} = \vec{y}G_d$, then we will have $\vec{x}A = \vec{z}B$ which means that $\vec{x} \in \psi(M')$ and hence $\phi(M') \leq \psi(M')$ for any finitely presented module M' . By Lemma 2.29, we can conclude that ϕ/ψ .

■
□

2.3 Quiver

Quiver is a useful example for understanding pp-condition, therefore we will introduce some basic definitions and concepts about quiver in this short section.

Definition 2.33

(Schiffler (2014) Definition 1.1)

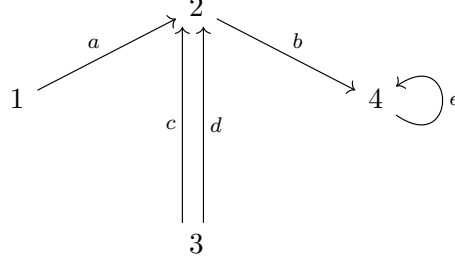
A quiver is a tuple $Q = (Q_p, Q_a, s, t)$ where Q_p is a set of vertices, Q_a is a set of arrows, $s, t : Q_a \rightarrow Q_p$ are maps sending arrow to its starting and terminal point respectively.

We will represent an element $\alpha \in Q_1$ as follows.

$$s(\alpha) \xrightarrow{\alpha} t(\alpha)$$

Example 2.34

Here is an example of quiver given by $Q_0 = \{1, 2, 3, 4\}$, $Q_1 = \{a, b, c, d, e\}$, $s(a) = 1$, $s(b) = 2$, $s(c) = 3 = s(d)$, $s(e) = 4$; $t(a) = 2$, $t(b) = 4 = t(e)$, $t(c) = 2 = t(d)$.



Definition 2.35

(Schiffler (2014) Definition 1.2)

A representation $M = (M_i, \phi_a)_{i \in Q_0, a \in Q_1}$ of a quiver Q is a collection of $\mathbb{K}$ -vector spaces M_i for each $i \in Q_0$, and a collection of $\mathbb{K}$ -linear maps $\phi_a : M_{s(a)} \rightarrow M_{t(a)}$ for each $a \in Q_1$.

A representation M is called finite-dimensional if each vector space M_i is finite-dimensional, and we call the vector $(\dim M_i)_{i \in Q_0}$ the dimension vector $\vec{\dim} M$. An element of a representation M is a tuple $(m_i)_{i \in Q_0}$ such that $m_i \in M_i$.

Example 2.36

(Schiffler (2014) Example 1.2)

Let Q be the quiver $1 \xrightarrow{a} 2$. Then

$$\begin{aligned} M & \quad \mathbb{K} \xrightarrow{1} \mathbb{K}, \\ M' & \quad \mathbb{K} \xrightarrow{0} \mathbb{K}, \\ M'' & \quad \mathbb{K}^2 \xrightarrow{\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}} \mathbb{K}^3 \end{aligned}$$

are representations of Q .

While quiver representations is defined over vector spaces, our emphasis on modules necessitates a broader framework. Therefore we adopt the generalized definition introduced by Mike Prest.

Definition 2.37

(Prest (2009) p.541)

Let R be any ring and Q be a quiver. An R -representation of Q , $M = (M_i, \phi_a)_{i \in Q_0, a \in Q_1}$, is a collection of R -module M_i for each $i \in Q_0$, and a collection of R -linear maps $\phi_a : M_{s(a)} \rightarrow M_{t(a)}$ for each $a \in Q_1$.

Definition 2.38

(Schiffler (2014) Definition 2.1)

Let $Q = (Q_0, Q_1, s, t)$ be a quiver with $i, j \in Q_0$. A path from i to j of length l in Q is a sequence

$$c = (i \mid a_1, a_2, \dots, a_l \mid j),$$

with $a_h \in Q_1$ such that $s(a_1) = i$, $s(a_h) = t(a_{h-1})$ for $2 \leq h \leq l$, $t(a_l) = j$.

We define $S(c) = s(a_1) = i$, and $T(c) = t(a_l) = j$. Notice that S and T is capitalised to prevent confusion between the source and target function on arrows.

Remark 2.39

Every path mentioned in this essay is finite, namely each path have a length n for some finite number n .

Example 2.40

- (Schiffler (2014) Example 2.2)
The constant path, or the trivial path at vertex i is any path from i to i of length 0. Explicitly, $(i \parallel i)$. This is denoted as e_i .
- Each arrow a is a path of length 1 from $s(a)$ to $t(a)$.

Definition 2.41

(Prest (2009) p.542)

Let Q be a quiver, R be any ring. The path algebra RQ of Q is a free R -module generate by the set of all finite paths with multiplication defined as follows.

For any two paths $p = (i_p \mid a_1, \dots, a_n \mid j_p)$ and $q = (i_q \mid b_1, \dots, b_m \mid j_q)$.

$$pq = \begin{cases} (i_p \mid a_1, \dots, a_n, b_1, \dots, b_m \mid j_q) & \text{if } i_q = S(q) = T(p) = j_p \\ 0 & \text{otherwise} \end{cases}$$

Remark 2.42

Notice that any R -representation of Q , $M = (M_i, \phi_a)_{i \in Q_0, a \in Q_1}$, can be consider as module over the path algebra RQ . Consider $M' = \bigoplus_{i \in Q_0} M_i$, we can

define an action of RQ on M' by the following.

- For any $\vec{m} = (m_i)_i \in M'$ and for any constant path e_i for $i \in Q_0$, we have

$$\vec{m}e_i := (\delta_{ij}m_j)_j,$$

where $\delta_{ij} = 0$ for any $j \neq i$ and $\delta_{ii} = 1$.

- For any arrow $a : i \rightarrow j$,

$$\vec{m}a := (n_k)_{k \in Q_0},$$

where $n_k = 0$ for any $k \neq j$ and $n_j = \phi_a(m_i)$.

- For any path $p = (i \mid a_1, \dots, a_n \mid j)$, we can define recursively that,

$$\vec{m}p := (\dots((\vec{m}a_1)a_2)\dots a_n).$$

Since this modules is completely determined by the collection of modules $(M_i)_{i \in Q_0}$ and R -morphisms $(\phi_a)_{a \in Q_1}$, we can see that any R -representation uniquely corresponds to such a modules up to isomorphisms. Thus any R -representation can be considered as a module over the path algebra over RQ in this sense.

Example 2.43

Consider a simple quiver Q :

$$1 \xrightarrow{a} 2 \xrightarrow{b} 3.$$

Let $M = (\mathbb{R}, \mathbb{R}, \mathbb{R}, \phi_a = f, \phi_b = g)$ be a $\mathbb{R}$ -representation as follows:

$$\mathbb{R} \xrightarrow{f} \mathbb{R} \xrightarrow{g} \mathbb{R}.$$

Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = 2x$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = x + 1$. Consider $(3, \pi, 110) \in \bigoplus_{1 \leq i \leq 3} \mathbb{R}$, we have the following,

$$(3, \pi, 110)e_1 = (3, 0, 0),$$

$$(3, \pi, 110)e_2 = (0, \pi, 0),$$

$$(3, \pi, 110)a = (0, f(3), 0) = (0, 6, 0).$$

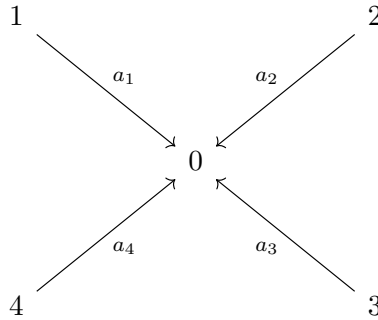
Let $p = (1 \mid a, b \mid 3)$, we have,

$$(3, \pi, 110)p = ((3, \pi, 110)a)b = (0, f(3), 0)b = (0, 0, g \circ f(3)) = (0, 0, 7).$$

Example 2.44

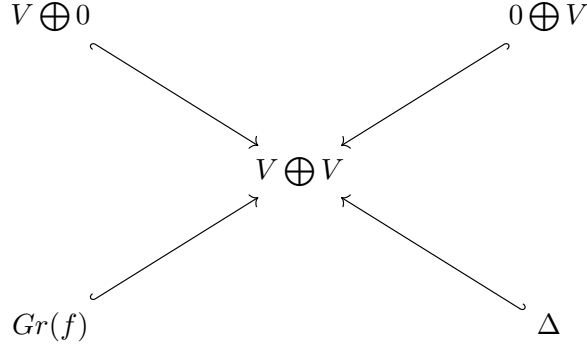
(Prest (2009) Example 1.1.3)

Let Q be the quiver $\tilde{D}_4$ which is as shown.



Let $\mathbb{K}$ is any field and V is an $\mathbb{K}$ -vector space, let $f : V \rightarrow V$ be an endomorphism. We can implement the structure (V, f) in a $\mathbb{K}$ -representation of $\tilde{D}_4$, called V_r , as follows.

Let $V_0 = V \oplus V$, $V_1 = V \oplus 0$, $V_2 = 0 \oplus V$, $V_3 = \Delta = \{(a, a) : a \in V\}$, $V_4 = Gr(f) = \{(a, f(a)) : a \in V\}$. Let a_1, a_2, a_3, a_4 be the inclusions of the subspaces respectively.



Consider the representation V_r as module over $\mathbb{K}\tilde{D}_4$, denoted as V'_r , defined by the way mentioned in Remark 2.42. Any element $\vec{a} \in V'_r$ is of the form $\vec{a} = ((v_1, v_2), (v_3, 0), (0, v_4), (v_5, v_5), (v_6, f(v_6)))$, where $v_i \in V$ for $1 \leq i \leq 6$.

Let $\eta_1(\vec{x}) : \vec{x} = \vec{x}e_1$ be a pp-condition. Then any $\vec{z} \in \eta_1(V'_r)$ is of the form $\vec{z} = (0, (z', 0), 0, 0, 0)$ for some $z' \in V$. Similarly, let $\eta_2(\vec{x}) : \vec{x} = \vec{x}e_2$ be a pp-condition, then any $\vec{u}_2 \in \eta_2(V'_r)$ is of the form $\vec{u}_2 = (0, 0, (0, u'_2), 0, 0)$ for some $u'_2 \in V$. Now consider the pp-condition,

$$\eta_3(\vec{x}) : \vec{x} = \vec{x}e_1 \wedge \exists \vec{u}_2, \vec{u}_4 \left(\vec{u}_2 = \vec{u}_2e_2 \wedge \vec{u}_4 = \vec{u}_4e_4 \wedge \vec{x}a_1 + \vec{u}_2a_2 = \vec{u}_4a_4 \right).$$

Explicitly, any $\vec{z}$ satisfies this pp-condition if $\vec{z} = (0, (z', 0), 0, 0, 0)$ with $z' \in V$ and there exists some $u'_2, u'_4 \in V$ such that $((z', 0) + (0, u'_2), 0, 0, 0, 0) = ((u'_4, f(u'_4)), 0, 0, 0, 0, 0)$. In other words, $z' = u'_4$ and $u'_2 = f(u'_4)$. Notice that since all the a_i are inclusion maps, hence it only results as shifting components to another position. Verbally, this pp-condition is to find some z' that has its image under f in V , and because f is an endomorphisms of V , therefore any $z' \in V$ satisfies such condition.

Consider the pp-condition $\eta(\vec{x}, \vec{y})$,

$$\begin{aligned}
 \vec{x} = \vec{x}e_1 \wedge \vec{y} = \vec{y}e_1 \wedge \exists \vec{u}_2, \vec{u}_4, \vec{u}_3 \left(\vec{u}_2 = \vec{u}_2e_2 \wedge \vec{u}_4 = \vec{u}_4e_4 \wedge \vec{x}a_1 + \vec{u}_2a_2 = \vec{u}_4a_4 \right. \\
 \left. \wedge \vec{u}_3 = \vec{u}_3e_3 \wedge \vec{y}a_1 + \vec{u}_2a_2 = \vec{u}_3a_3 \right).
 \end{aligned}$$

By the previous paragraph, we can see that $\vec{x}, \vec{y}$ is determined by $x', y' \in V$ respectively by $\vec{x} = (0, (x', 0), 0, 0, 0)$, $\vec{y} = (0, (y', 0), 0, 0, 0)$. Unpacking this condition, one can check that $(\vec{x}, \vec{y}) \in \eta(V_r')$ if and only if $x' \in V$ and $y' = f(x')$.

2.4 Finitely presented functors

In this section, we will introduce the category of pp-pairs and established several lemmas that together yields the final theorem.

Definition 2.45

(Prest (2009) Section 3.2.2) Let $\mathbb{L}_R^{eq+}$ denote the category of pp-pairs. The objects of this category are the pp-pairs ϕ/ψ ; a morphism from ϕ/ψ to the pair ϕ'/ψ' is given by a pp-condition $\rho(\vec{x}, \vec{x}')$ which satisfies:

- $\rho(\vec{a}, \vec{a}')$ and $\phi(\vec{a})$ both holds implies that $\phi'(\vec{a}')$ holds.
- $\rho(\vec{a}, \vec{a}')$ and $\psi(\vec{a})$ both holds implies that $\psi'(\vec{a}')$ holds.
- $\phi(\vec{a})$ holds implies that there exists $\vec{a}'$ such that $\rho(\vec{a}, \vec{a}')$ holds.

Remark 2.46

Observe that we require

$$\phi(M_1)/\psi(M_1) = \phi(M_2)/\psi(M_2)$$

to hold for any module M_1, M_2 in order to have a nice definition of the category of pp-pairs, this result is given by the lemma below.

Lemma 2.47

(Prest (2009) Lemma 3.2.11) Let $M \in \text{ob}(\mathbf{Mod}\text{-}R)$ be a module, the evaluation at M is a functor from $ev_M : \mathbb{L}_R^{eq+} \rightarrow \mathbf{Ab}$ defined by:

- $\phi/\psi \mapsto \phi(M)/\psi(M)$.
- $\rho \mapsto \rho(M)$.

The evaluation at M , ev_M , is an exact (additive) functor for any module M .

Proof.

Claim: Any short exact sequence in $\mathbb{L}_R^{eq+}$ is isomorphic to

$$0 \rightarrow \phi'/\psi \rightarrow \phi/\psi \rightarrow \phi/\phi' \rightarrow 0,$$

for some $\psi \leq \phi' \leq \phi$.

The proof of the claim is provided at the end of the proof; we first prove the lemma with the assumption that the claim is true. Let M be any module, and let ev_M denote the evaluation at M . Consider the sequence in the claim and the following,

$$0 \rightarrow ev_M(\phi'/\psi) \rightarrow ev_M(\phi/\psi) \rightarrow ev_M(\phi/\phi') \rightarrow 0,$$

which is simply,

$$0 \rightarrow \phi'(M)/\psi(M) \rightarrow \phi(M)/\psi(M) \rightarrow \phi(M)/\phi'(M) \rightarrow 0.$$

Notice that we have $\psi(M) \subseteq \phi'(M) \subseteq \phi(M)$ by $\psi \leq \phi' \leq \phi$. Hence by the Lemma 1.57, we can concluded that this is an exact sequence and ev_M is an exact functor.

Finally, we need to look back and prove the claim that is asserted in the beginning.

Proof of the claim: Consider a short exact sequence in $\mathbb{L}_R^{eq+}$

$$0 \rightarrow \phi_1/\psi_1 \xrightarrow{\rho_1} \phi_2/\psi_2 \xrightarrow{\rho_2} \phi_3/\psi_3 \rightarrow 0,$$

for $\phi_i \leq \psi_i, i = 1, 2, 3$.

Define ϕ' as the pp-condition $\exists \vec{x} \rho_1(\vec{x}, \vec{y})$. Notably, this is the image of ρ_1 and thus $\phi' \leq \phi_2$, similarly $\psi_2 \leq \phi'$ as the image of ρ_1 must also contain the kernel of ρ_1 . Hence we have the inequality $\psi_2 \leq \phi' \leq \phi_2$.

Define $\rho'_2 : \phi_2/\phi' \rightarrow \phi_3/\psi_3$ as $\rho'_2(\vec{x}, \vec{y}) \iff \rho_2(\vec{x}, \vec{y})$ and $\rho'_1 : \phi_1/\psi_1 \rightarrow \phi'/\psi_2$ as $\rho'_1(\vec{x}, \vec{y}) \iff \rho_1(\vec{x}, \vec{y})$ for any tuple $\vec{x}, \vec{y}$.

Define $\rho''_2 : \phi_3/\psi_3 \rightarrow \phi_2/\phi'$ as $\rho''_2(\vec{y}, \vec{x}) \iff \rho_2(\vec{y}, \vec{x})$ and $\rho''_1 : \phi_1/\psi_1 \rightarrow \phi'/\psi_2$ as $\rho''_1(\vec{x}, \vec{y}) \iff \rho_1(\vec{y}, \vec{x})$ for any tuple $\vec{x}, \vec{y}$.

These definition are well defined by the inequality mentioned above. Consider the following diagram,

$$\begin{array}{ccccccc} 0 & \longrightarrow & \phi_1/\psi_1 & \xrightarrow{\rho_1} & \phi_2/\psi_2 & \xrightarrow{\rho_2} & \phi_3/\psi_3 \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \uparrow \\ & & \rho'_1 & & 1_{\phi_2/\psi_2} & & \rho'_2 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \phi'/\psi_2 & \xrightarrow{1_{\phi_2/\psi_2} \circ \rho_1 \circ \rho''_1} & \phi_2/\psi_2 & \xrightarrow{\rho'_1 \circ \rho_2 \circ 1_{\phi_2/\psi_2}} & \phi_2/\phi' \longrightarrow 0 \end{array}$$

The exactness of the lower sequence is given by the inequality $\psi_2 \leq \phi' \leq \phi_2$ and Lemma 1.57. One can check that the ρ'_i, ρ''_i are isomorphisms for $i = 1, 2$, as $\rho'_i(\vec{x}, \vec{y}) \iff \rho''_i(\vec{y}, \vec{x})$, therefore the morphism sends $[\vec{x}] \xrightarrow{\rho'_i}$

$[\vec{y}] \xrightarrow{\rho'_i} [\vec{x}]$ where $[\vec{x}]$ denotes the equivalence class of $\vec{x}$. Thus the short exact sequence is isomorphic. ■

□

Lemma 2.48

(Prest (2009) Lemma 3.2.10) $\mathbb{L}_R^{eq+}$ is an abelian category

The proof of this lemma is omitted as one can obtain a proof following from the last results of this essay, Theorem 2.51.

Lemma 2.49

(Prest (2009) Lemma 10.2.1)

Recall from Definition 1.26 the definition of H^f for morphism f .

- Let $F \in (\mathbf{Mod}\text{-}R, \mathbf{Ab})^{fp}$, then $F \cong \text{coker}(H^f)$ for some $f \in \mathbf{Mod}\text{-}R(A, B)$ and some $A, B \in \text{ob}(\mathbf{Mod}\text{-}R)$.
- Conversely, $\text{coker}(H^g)$ for some $g \in \mathbf{Mod}\text{-}R(C, D)$ and $C, D \in \text{ob}(\mathbf{Mod}\text{-}R)$ is finitely presented.

The proof of this lemma is beyond the scope of this essay.

In the following passage, we will start to prove that this category $\mathbb{L}_R^{eq+}$ is equivalent to the category of finitely presented functors from $\mathbf{Mod}\text{-}R$ to abelian groups, $(\mathbf{Mod}\text{-}R, \mathbf{Ab})^{fp}$ if R is any ring (or more generally any skeletally small preadditive category). This will help us to investigate problems in the area of logic (pp-pairs) from a categorical algebra perspective.

Lemma 2.50

(Prest (2009) Theorem 10.2.30) Every finitely presented functor from $\mathbf{Mod}\text{-}R$ to abelian groups, $F \in (\mathbf{Mod}\text{-}R, \mathbf{Ab})^{fp}$, is naturally isomorphic to some pp functor $F_{\phi/\psi}$.

Proof.

Let $F \in (\mathbf{Mod}\text{-}R, \mathbf{Ab})^{fp}$. Then we can find a morphism $f \in \mathbf{Mod}\text{-}R(A, B)$ such that $F \cong \text{coker}(H^f)$ by Lemma 2.49. Consider the sequence,

$$H^B \xrightarrow{H^f} H^A \rightarrow F \rightarrow 0,$$

where the arrow from H^A to F is the isomorphism from $\text{coker}(H^f)$ composite with the canonical morphism of the cokernel. Notice that this sequence is exact by construction.

Let $\vec{a}$ be a generating tuple of A , by Lemma 2.32 we have that $pp^A(\vec{a}) = \langle \phi_A \rangle$ for some pp-condition ϕ_A , similarly let $pp^B(f(\vec{a})) = \langle \phi_B \rangle$. By the isomorphism of given by Lemma 2.49, we have that,

$$F(M) \cong \text{coker}(H^f)(M) = H^A(M)/\text{im}(H^f(M)) = (A, M)/\{g \circ f : B \xrightarrow{g} M\},$$

which is the abelian group of morphisms from A to M modulo through those which factor along f .

Consider the functor F_{ϕ_A/ϕ_B} , $F_{\phi_A/\phi_B}(M) = \phi_A(M)/\phi_B(M)$. Observe that by Corollary 2.28, we have that for a tuple $\vec{c}$, $\vec{c} \in \phi_A(M)$ if and only if there exists a morphism $A \xrightarrow{h} M$ such that $h(\vec{a}) = \vec{c}$. This correspondence is unique because the image of $\vec{a}$ under a morphism is unique, namely for any $\vec{c}, \vec{d} \in \phi_A(M)$ with $h(\vec{a}) = \vec{c}$ and $h'(\vec{a}) = \vec{d}$, we have $h \neq h'$ because of $h(\vec{a}) \neq h'(\vec{a})$. Hence we can conclude that:

$$\phi_A(M) = \{ p(\vec{a}) : A \xrightarrow{p} M \}.$$

Analogously, Corollary 2.28 also shows that $\phi_B(M) = \{ q(f(\vec{a})) : B \xrightarrow{q} M \}$ by similar argument. Therefore, the following equation is true:

$$F_{\phi_A/\phi_B}(M) = \phi_A(M)/\phi_B(M) \cong \{ p(\vec{a}) : A \xrightarrow{p} M \} / \{ q(f(\vec{a})) : B \xrightarrow{q} M \}.$$

Construct an morphism $\eta_M : F(M) \rightarrow F_{\phi_A/\phi_B}(M)$ by evaluation mapping:

$$k' \mapsto k'(\vec{a}),$$

for any $k \in F(M)$ and $k' \in (A, M)/\{ g \circ f : B \xrightarrow{g} M \}$ is what k maps to in the isomorphism. Note that η_M is an isomorphism because $\vec{a}$ is the generating tuple of M , so the image $k(\vec{a})$ determines k , hence it is injective. Surjectivity of the morphism follows directly from the definition. The following diagram commutes:

$$\begin{array}{ccc} F(M_1) & \xrightarrow{F(f)} & F(M_2) \\ \eta_{M_1} \updownarrow & & \updownarrow \eta_{M_2} \\ F_{\phi_A/\phi_B}(M_1) & \xrightarrow{F_{\phi_A/\phi_B}(f)} & F_{\phi_A/\phi_B}(M_2) \end{array}$$

Thus the collection of these isomorphism $(\eta_M)_{M \in \text{ob}(\mathbf{Mod}\text{-}R)}$ is a natural isomorphism $F \cong F_{\phi_A/\phi_B}$ by Lemma 1.17. $\square$

Theorem 2.51

(Prest (2009) Theorem 10.2.30) Let R be a skeletally small preadditive category. Then the category of finitely presented functors from $\mathbf{Mod}\text{-}R$ to abelian groups, $(\mathbf{Mod}\text{-}R, \mathbf{Ab})^{fp}$, is equivalent to the category of pp-pairs, $\mathbb{L}_R^{eq+}$.

Proof.

Prest (2009)

Firstly, we have already shown the equivalence on object by Lemma 2.50. It suffices to show that any morphisms between pp-pairs correspond to a natural transformation between the corresponding functors.

Let $\rho : \phi/\psi \rightarrow \phi'/\psi'$ be a morphism in $\mathbb{L}_R^{eq+}$. For any finitely presented module M , consider the additive function $\tilde{\rho}_M : \phi(M)/\psi(M) \rightarrow \phi'(M)/\psi'(M)$ defined by $\vec{a} + \psi(M) \mapsto \vec{a}' + \psi'(M)$ such that $\vec{a}'$ is the unique tuple with $(\vec{a}, \vec{a}') \in \rho(M)$ up to modulo $\psi'(M)$. The existence of such tuple is given by the Definition 2.45. By Lemma 2.12, for any morphism $f : M \rightarrow M'$, we have $(f(\vec{a}), f(\vec{a}')) \in \rho(M')$. Hence, we have that the collection of all $\tilde{\rho}_M$ for $M \in \mathbf{Mod}\text{-}R$ define a natural transformation, denoted as $\tilde{\rho}$. Explicitly, the following diagram commutes.

$$\begin{array}{ccc}
\phi(M)/\psi(M) & \xrightarrow{F_{\phi/\psi}(f)} & \phi(M')/\psi(M') \\
\downarrow \tilde{\rho}_M & & \downarrow \tilde{\rho}_{M'} \\
\phi'(M)/\psi'(M) & \xrightarrow{F_{\phi'/\psi'}(f)} & \phi'(M')/\psi'(M')
\end{array}$$

Conversely, let $\tau : F_{\phi/\psi} \rightarrow F_{\phi'/\psi'}$ be a natural transformation. Consider its component at some finitely presented module C , namely τ_C . Let $\vec{c}$ and $\vec{c}'$ be tuple from C such that $(C, \vec{c})$ is the free realization of ϕ and $\vec{c}' + \psi'(C) = \tau_C(\vec{c} + \psi(C))$. Let $\eta(\vec{c}, \vec{c}')$ be a generator of $pp^C(\vec{c}, \vec{c}')$. In other words, $(C, (\vec{c}, \vec{c}'))$ is a free realization of η .

Claim: η defines a pp-morphism from ϕ/ψ to ϕ'/ψ' .

With the assumption, it is sufficient to prove that if the natural transformation τ is equivalent to $\tilde{\eta}$, given by the pp-morphism η , for η is the pp-morphism defined in the above paragraph.

Let $M \in \mathbf{Mod}\text{-}R$, $\vec{d} \in \phi(M)$, then by Lemma 2.28, there is a morphism $f : C \rightarrow M$ such that $f(\vec{c}) = \vec{d}$. By the naturality square,

$$\begin{array}{ccc}
\phi(C)/\psi(C) & \xrightarrow{F_{\phi/\psi}(f)} & \phi(M)/\psi(M) \\
\downarrow \tau_C & & \downarrow \tau_M \\
\phi'(C)/\psi'(C) & \xrightarrow{F_{\phi'/\psi'}(f)} & \phi'(M)/\psi'(M)
\end{array}$$

We have,

$$\tau_M \circ F_{\phi/\psi} f = F_{\phi'/\psi'} f \circ \tau_C. \quad (2)$$

Analogously, we have,

$$\tilde{\eta}_M \circ F_{\phi/\psi} f = F_{\phi'/\psi'} f \circ \tilde{\eta}_C. \quad (3)$$

Also consider that $\tau_C(\vec{c} + \psi(C)) = \vec{c}' + \psi'(C) = \tilde{\eta}_C(\vec{c} + \psi(C))$ since $(\vec{c}, \vec{c}') \in \eta(C)$ by definition. Therefore, we have

$$\begin{aligned} \tau_M(\vec{d} + \psi(M)) &= \tau_M(f(\vec{c}) + \psi(M)) \\ &= \tau_M F_{\phi/\psi}(f)(\vec{c} + \psi(C)) \\ &= F_{\phi'/\psi'}(f) \circ \tau_C(\vec{c} + \psi(C)) \end{aligned} \quad (2)$$

$$\begin{aligned} &= F_{\phi'/\psi'}(f) \circ \tilde{\eta}_C(\vec{c} + \psi(C)) \\ &= \tilde{\eta}_M \circ F_{\phi/\psi}(f)(\vec{c} + \psi(C)) \\ &= \tilde{\eta}_M(f(\vec{c}) + \psi(M)) \\ &= \tilde{\eta}_M(\vec{d} + \psi(M)). \end{aligned} \quad (3)$$

This proves that $\tau_M = \tilde{\eta}_M$ for any module M , namely $\tau = \eta$, the correspondence of pp-morphisms and natural transformations.

Proof of the claim: According to the Definition 2.45, we will need to show that η satisfy the three properties:

- $\eta(\vec{a}, \vec{a}')$ and $\phi(\vec{a})$ both holds implies that $\phi'(\vec{a}')$ holds.
- $\eta(\vec{a}, \vec{a}')$ and $\psi(\vec{a})$ both holds implies that $\psi'(\vec{a}')$ holds.
- $\phi(\vec{a})$ holds implies that there exists $\vec{a}'$ such that $\eta(\vec{a}, \vec{a}')$ holds.

Consider any finitely presented module M and tuple $(\vec{b}, \vec{b}') \in \eta(M)$. Notice that by Lemma 2.28, there is a morphism $f : C \rightarrow M$ such that $(\vec{c}, \vec{c}') \mapsto (\vec{b}, \vec{b}')$.

For the first condition, assume that $\vec{b} \in \phi(M)$. Then by the naturality square of τ we have,

$$F_{\phi'/\psi'} f \circ \tau_C(\vec{c} + \psi(C)) = F_{\phi'/\psi'} f(\vec{c}' + \psi'(C)),$$

which by the definition of $F_{\phi'/\psi'} f$ means that $F_{\phi'/\psi'} f(\vec{c}' + \psi'(C)) = \vec{b}' + \psi'(M) \in \phi'(M)/\psi'(M)$. Hence $\vec{b}' \in \phi'(M)$.

For the second condition, assume that $\vec{b} \in \psi(M)$. Since

$$\vec{b}' + \psi'(M) = \tau_M(\vec{b} + \psi(M)) = \tau_M(0 + \psi(M)) = 0 + \psi'(M),$$

therefore $\vec{b}' \in \psi'(M)$.

For the third condition, let $\vec{d} \in \phi(M)$ be a tuple. Using Corollary 2.28, there is a $C \xrightarrow{f'} M$ such that $\vec{c} \mapsto \vec{d}$. Denote $f'(\vec{c})$ as $\vec{e}$. Hence we have $f'((\vec{c}, \vec{c}')) = (\vec{d}, \vec{e})$. By the other direction of Corollary 2.28, it states that since we have a morphisms f' sending $(\vec{c}, \vec{c}')$ to $(\vec{d}, \vec{e})$, and $(C, (\vec{c}, \vec{c}'))$ is a free realization of η , therefore $(\vec{d}, \vec{e}) \in \eta(M)$. ■

□

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